NELSSA: A GPU-PNM Heterogeneous System for Mixed-Length LLM Serving via Length-based Request Placement



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Youngpyo Joo, Hoshik Kim, Jongse Park†

SK hynix Memory Systems Research
† **KAIST** CASYS Lab

Background: Mixed-Length Workloads



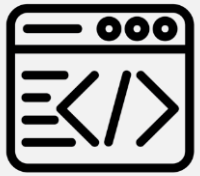
Conversational



Reasoning

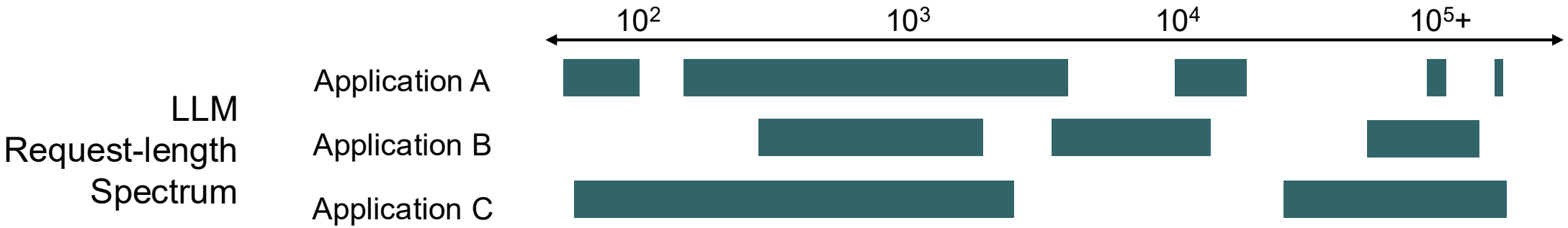


Multi-agent

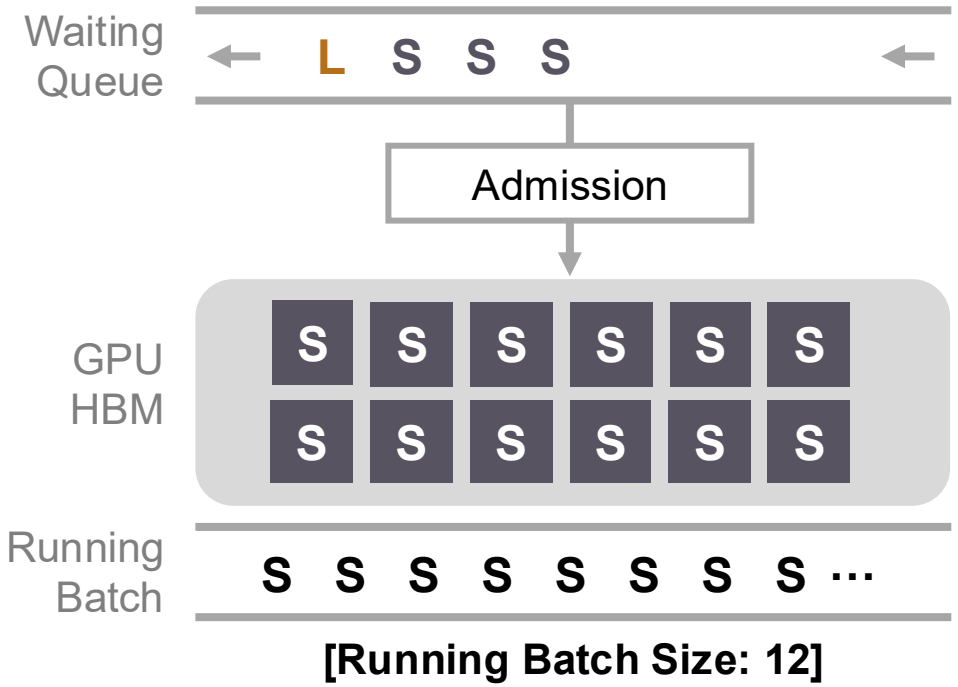


Coding

A wide and dynamic mix of short and long requests

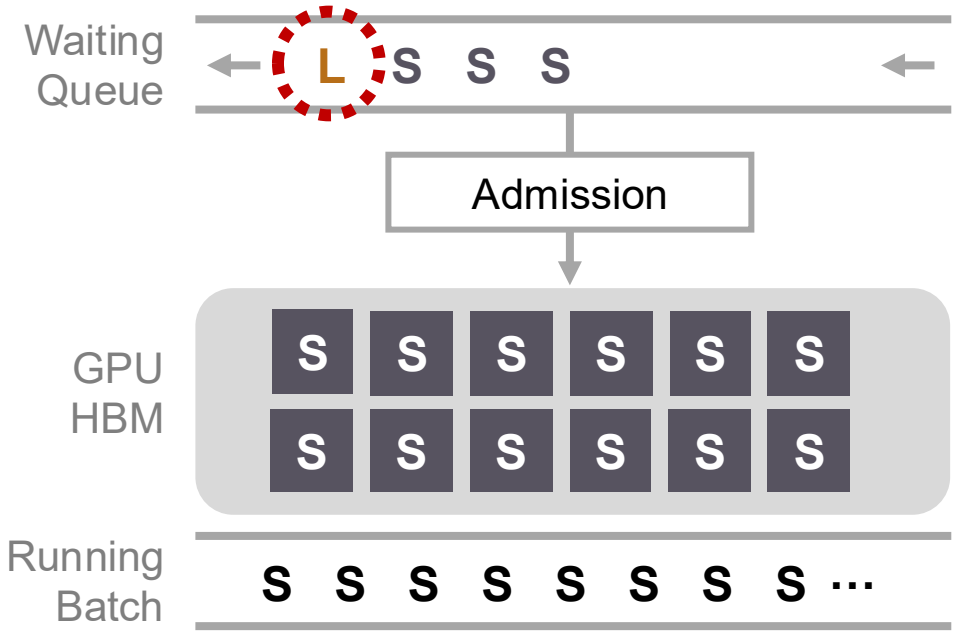


Motivation: Long-Context Interference



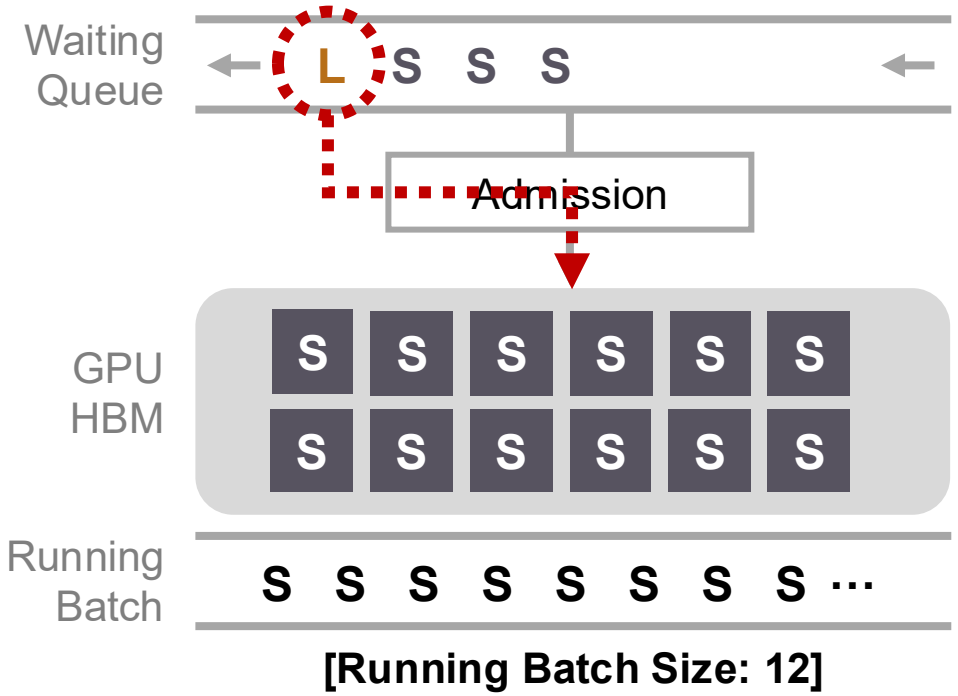
[Running Batch Size: 12]

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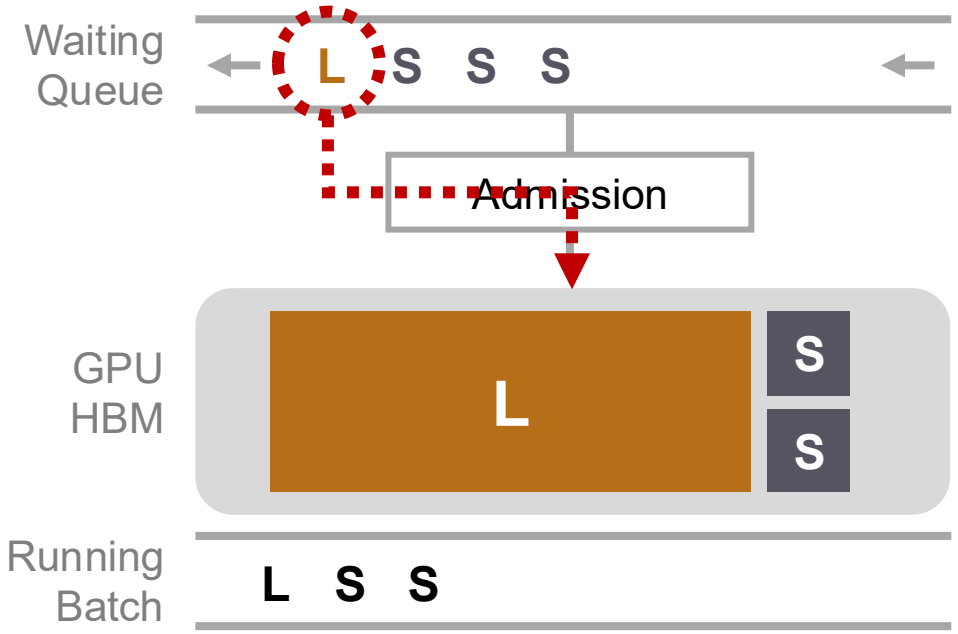
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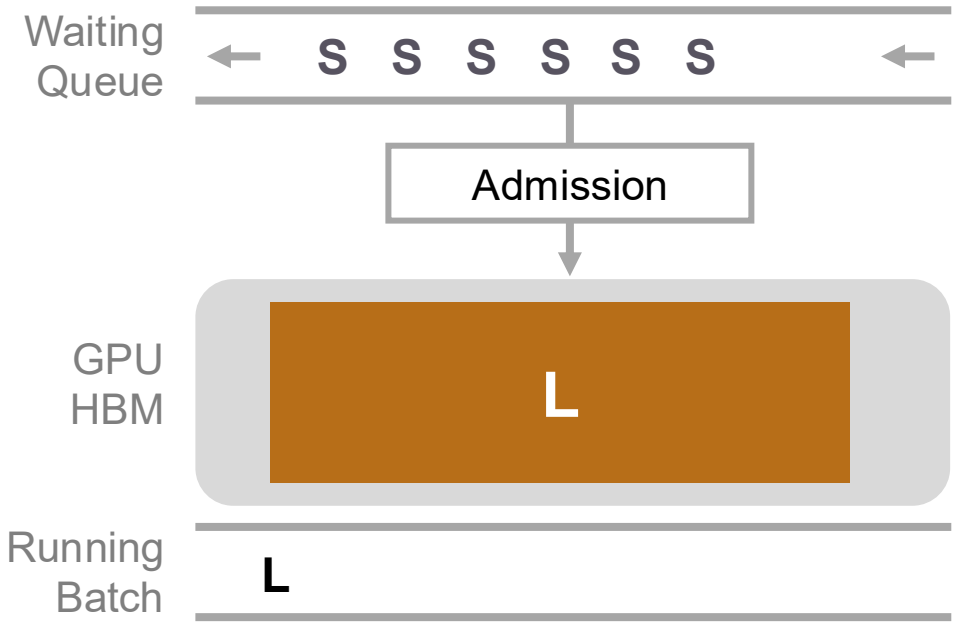
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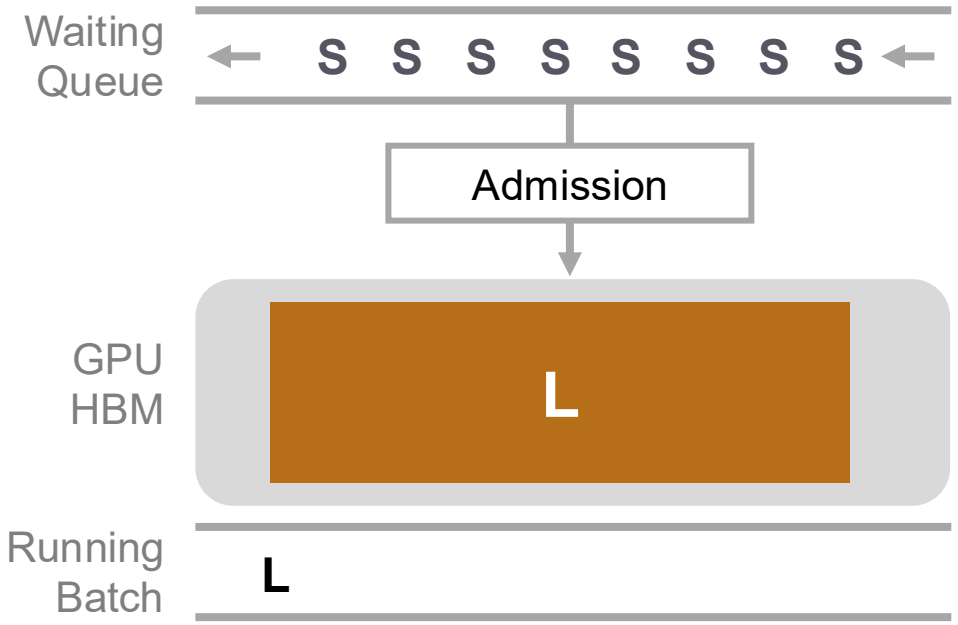
[Running Batch Size: 3]

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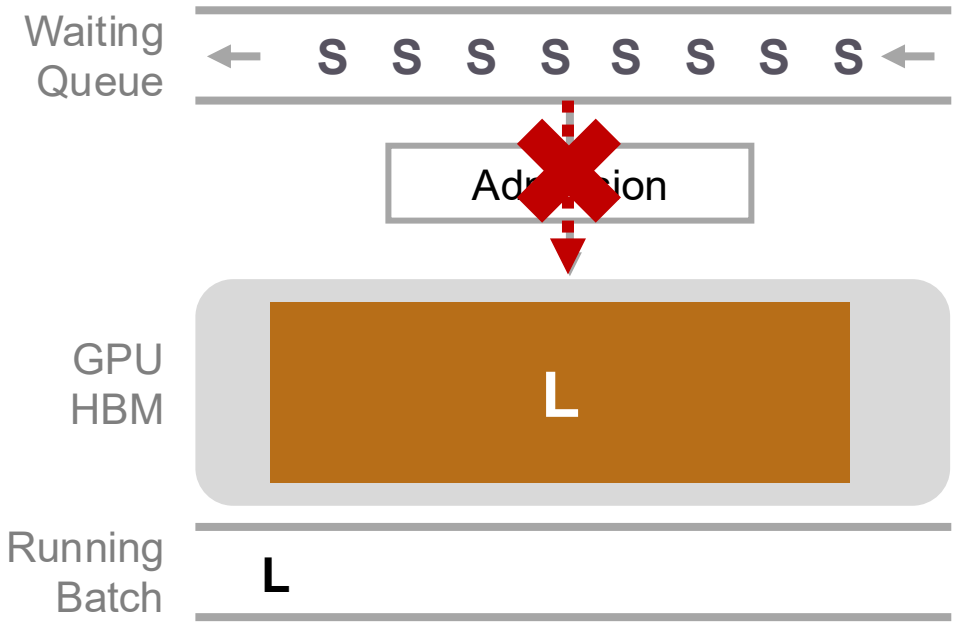
[Running Batch Size: 1]

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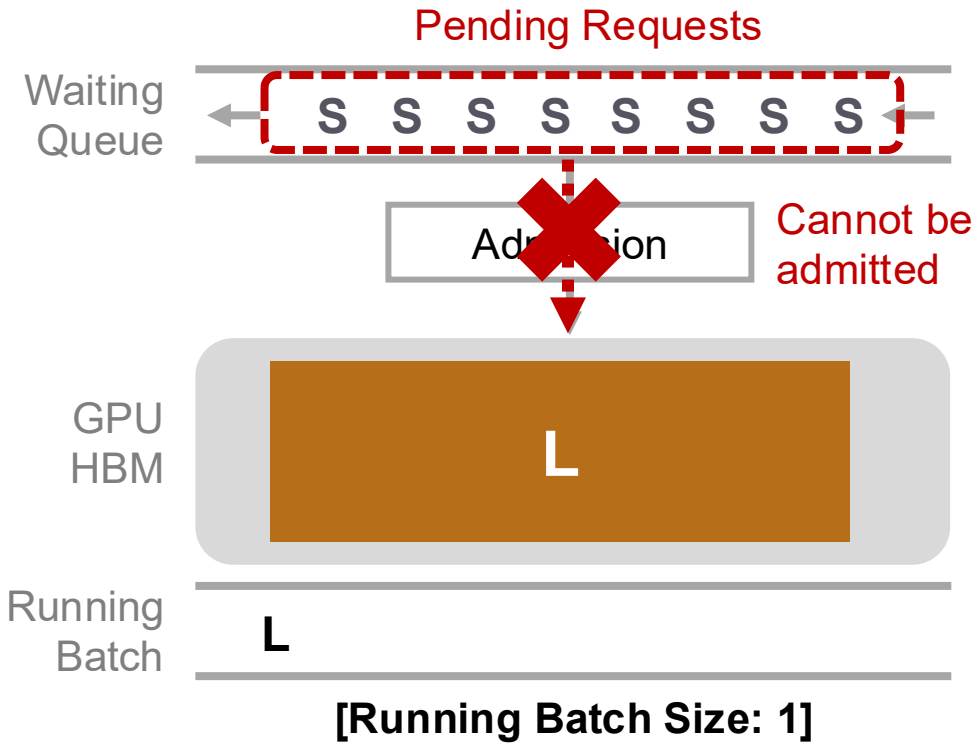
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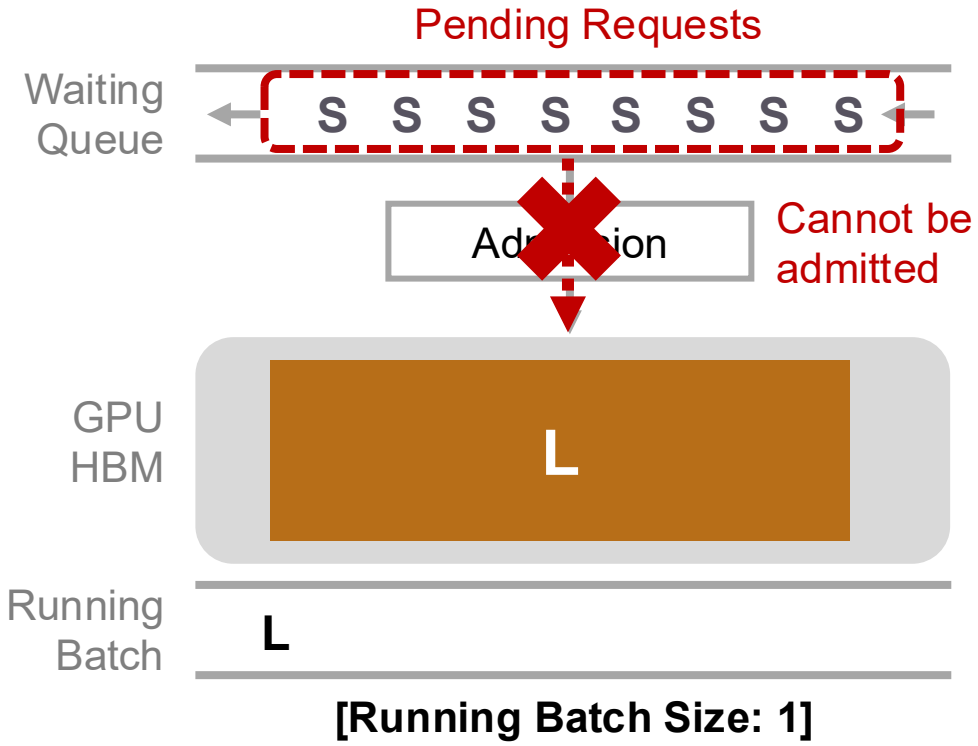
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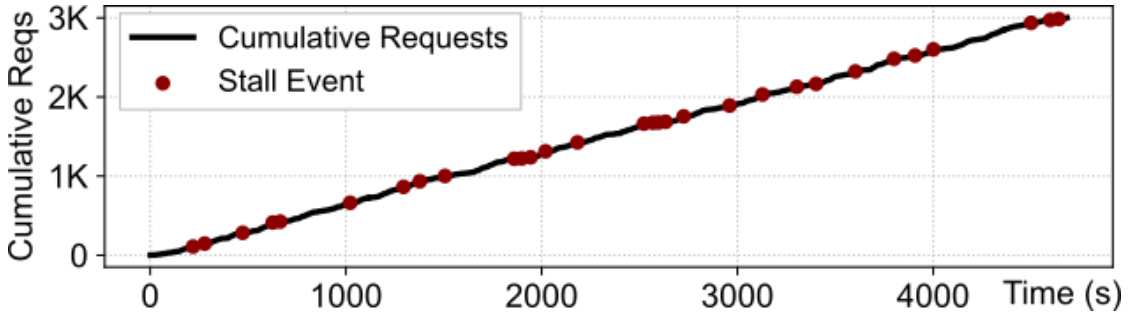
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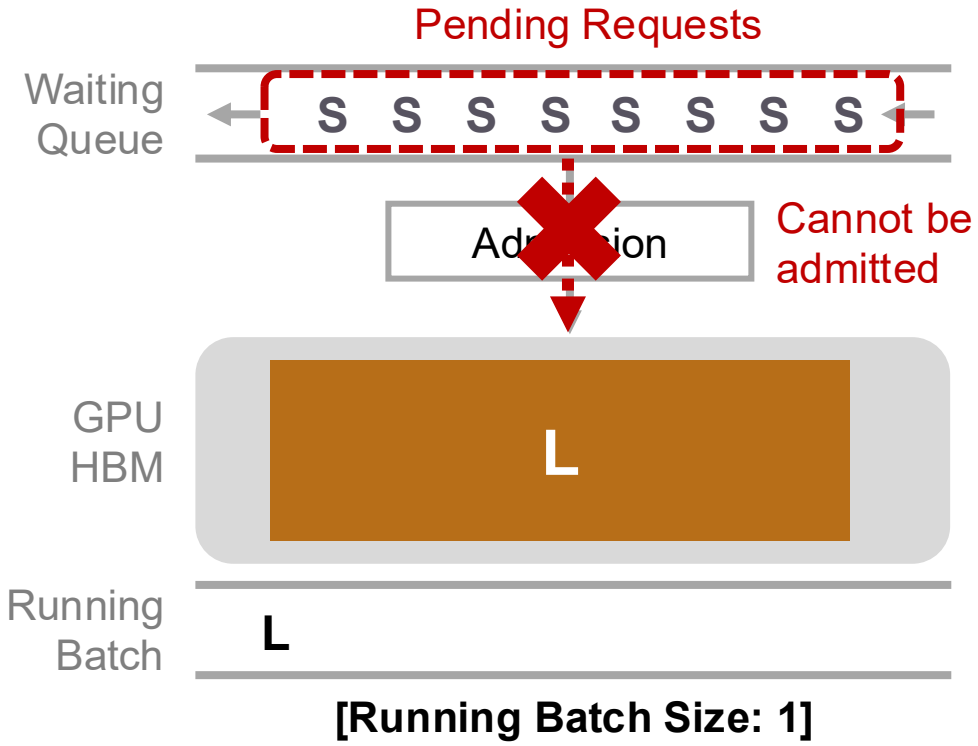
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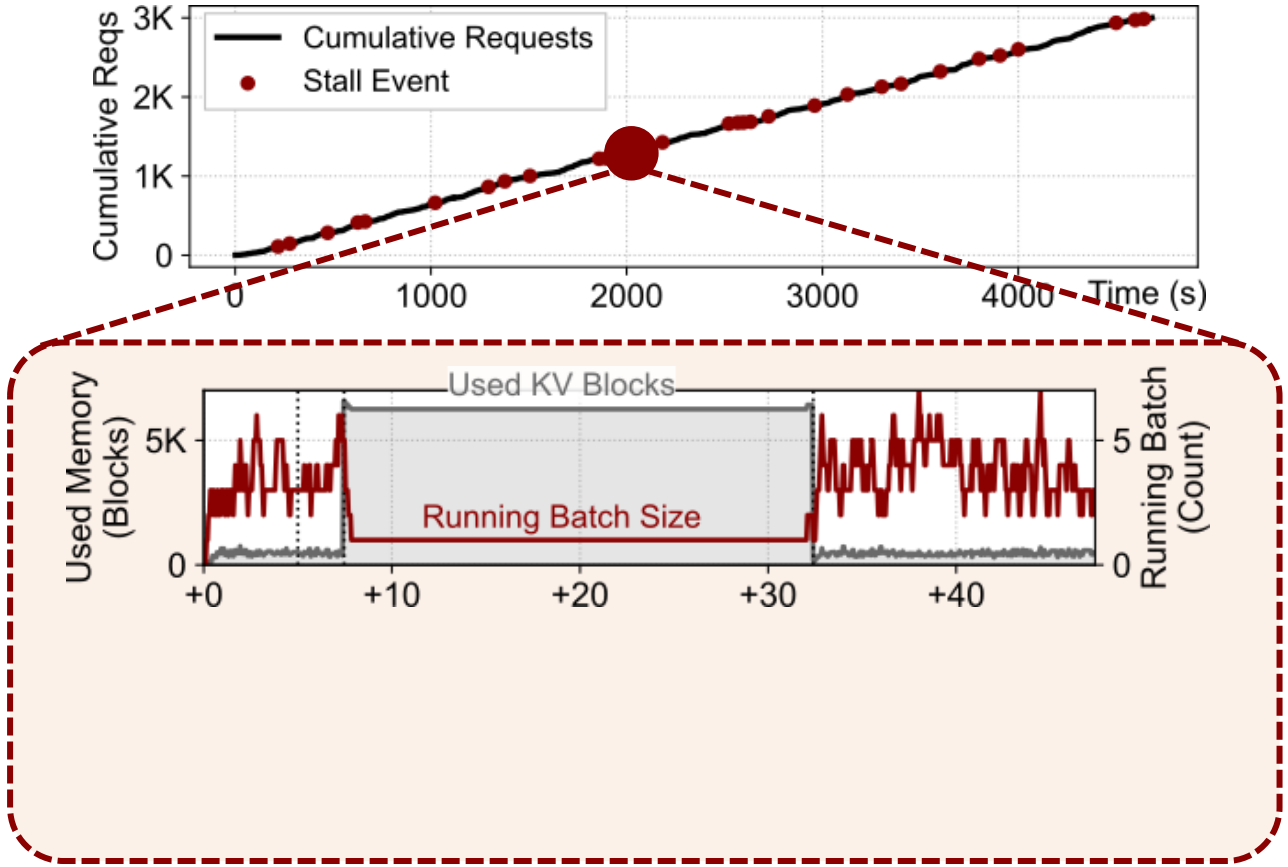
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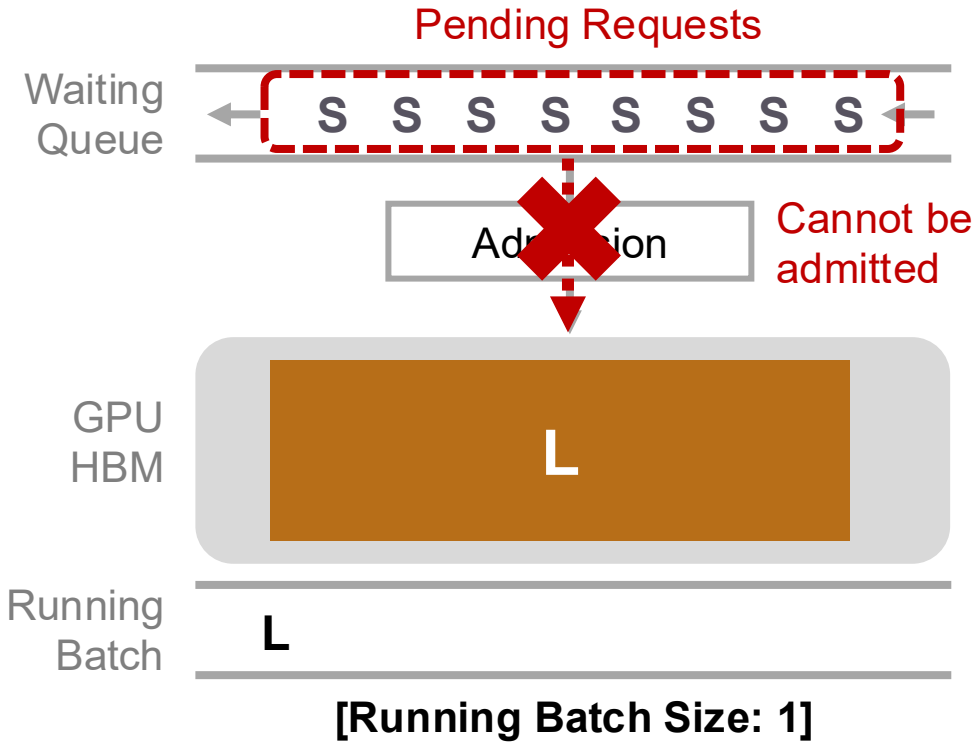


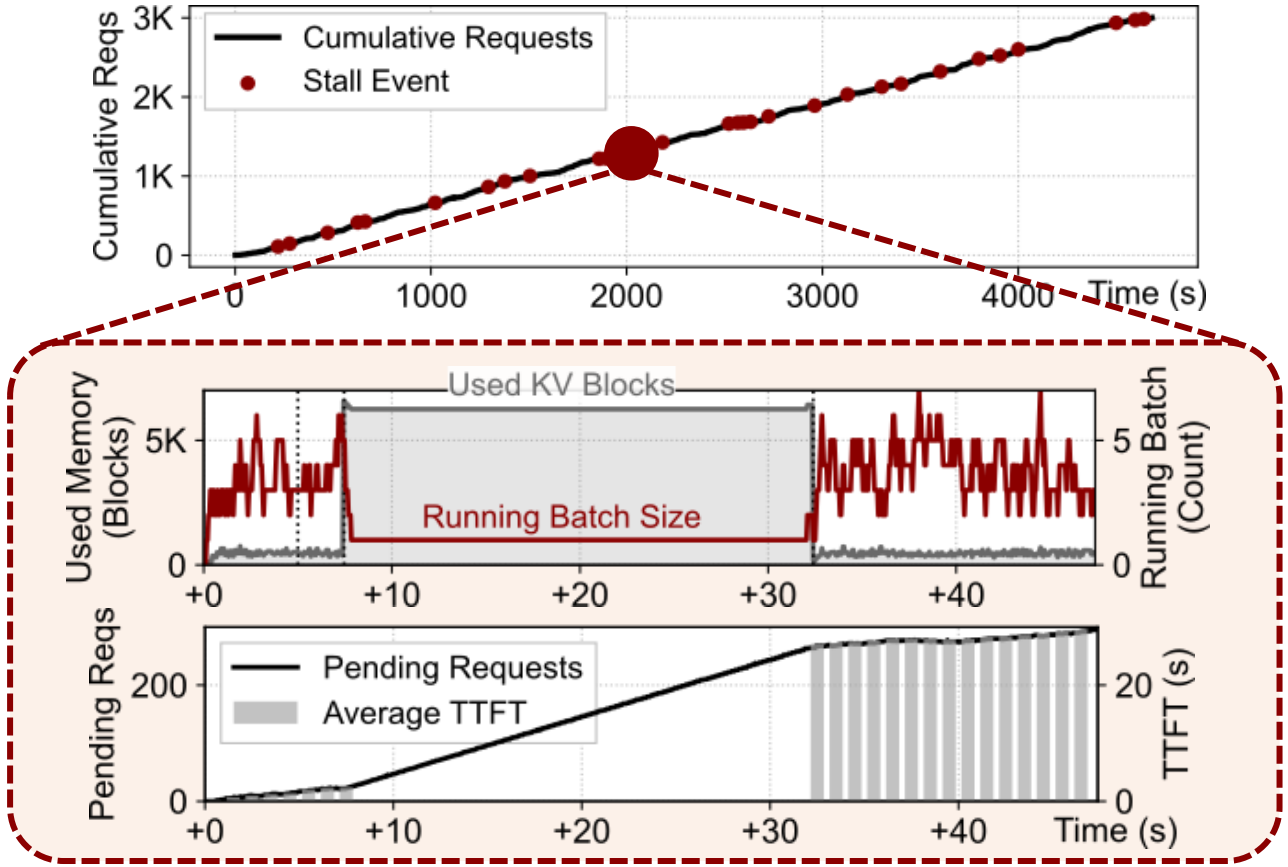
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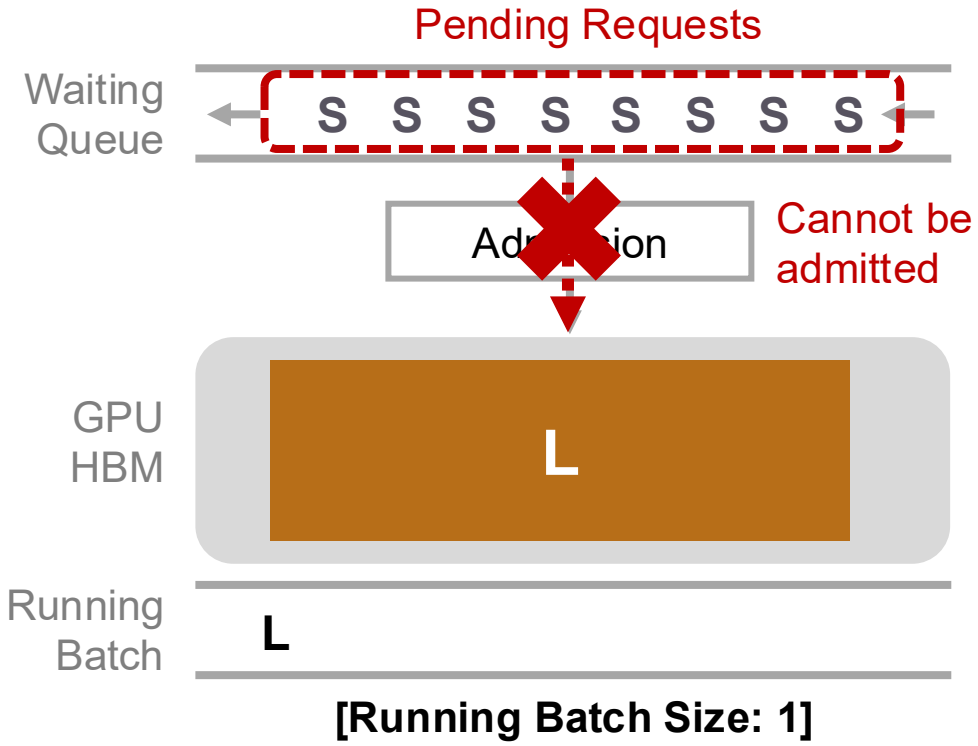


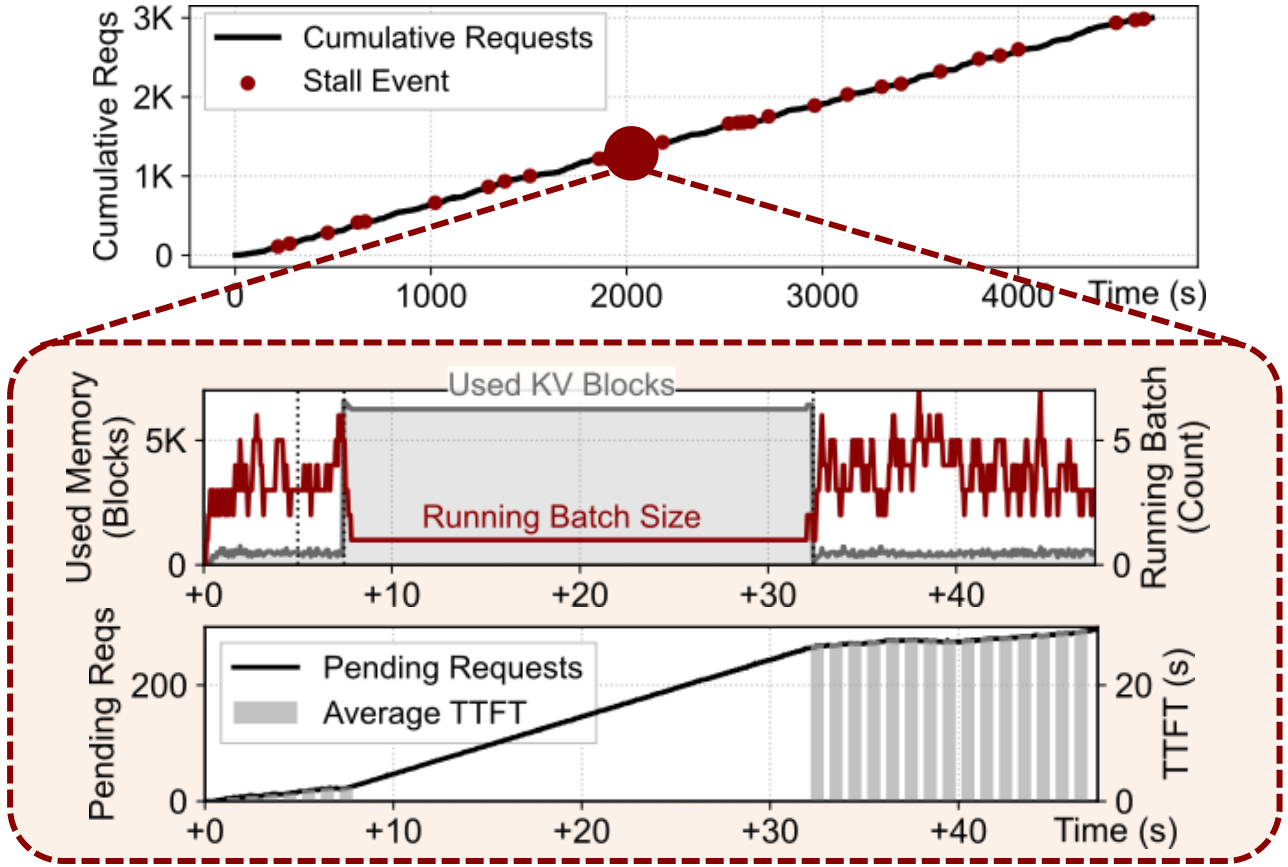
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Need to handle large KV state efficiently

Motivation: Existing Approaches

GPU Sparse Attention | KV Offloading | PNM Sparse Attention



Access only selected KV

- Quest ‘24
- DeepSeek Sparse Attention (DSA) ‘25
- Native Sparse Attention (NSA) ‘25
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\+ Reduced KV access
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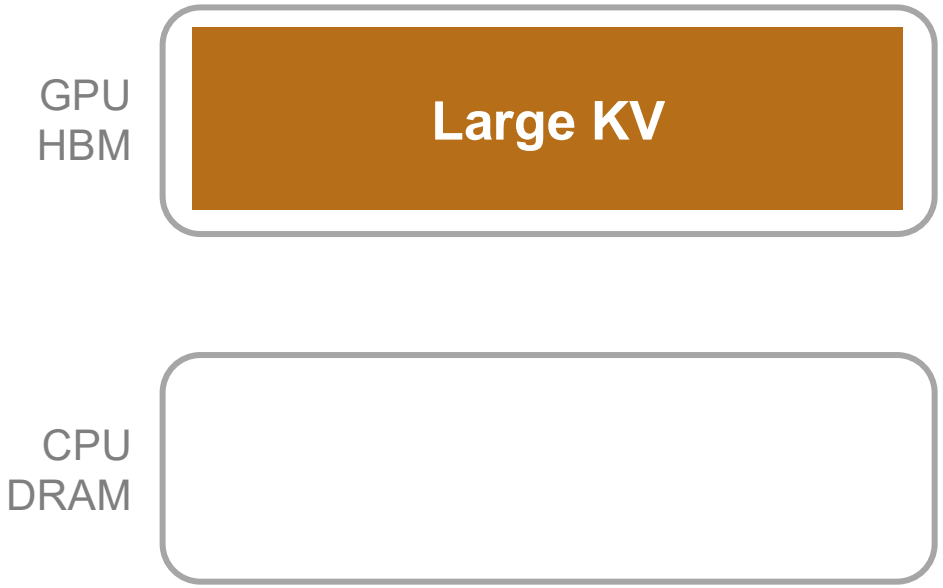
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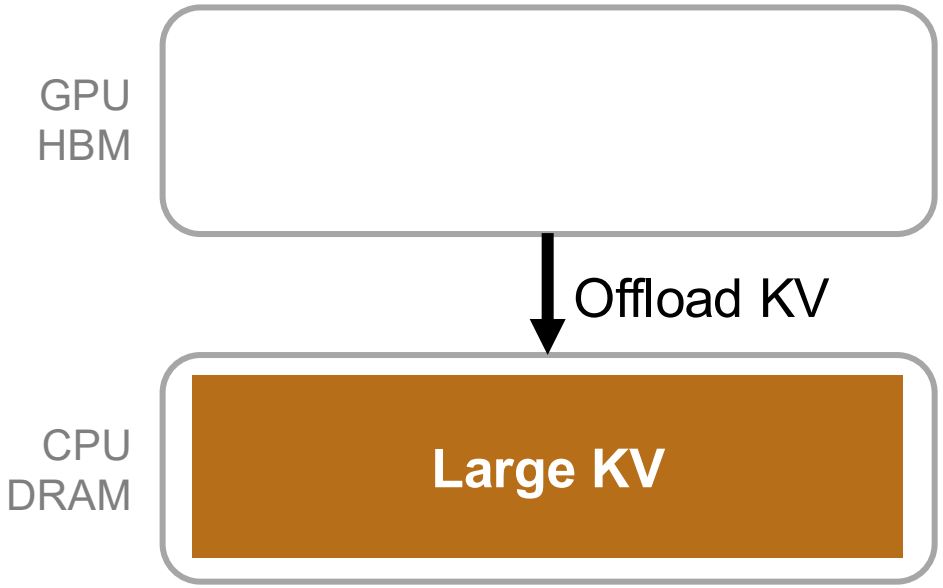
Move KV out of GPU HBM

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- HiSparse (’26)

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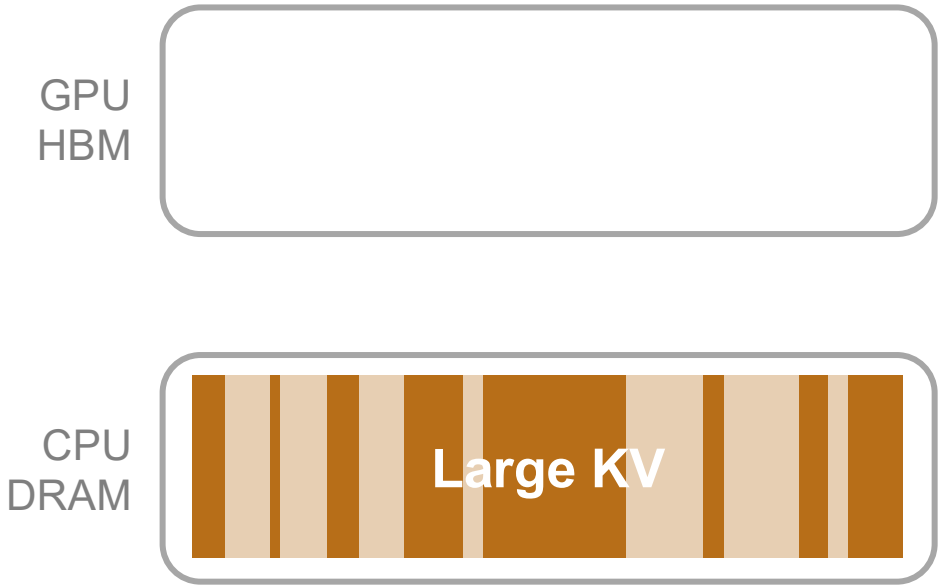
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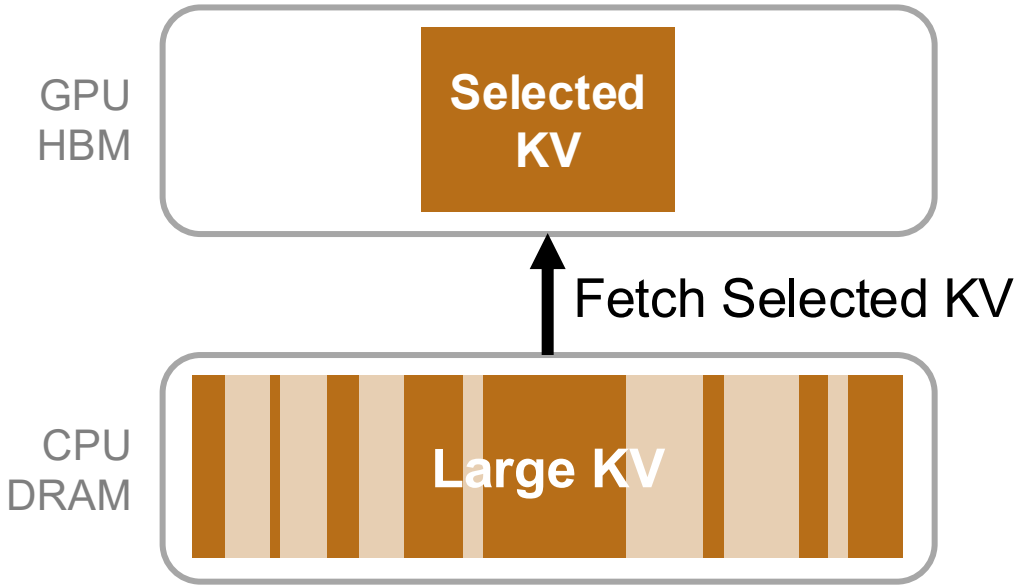
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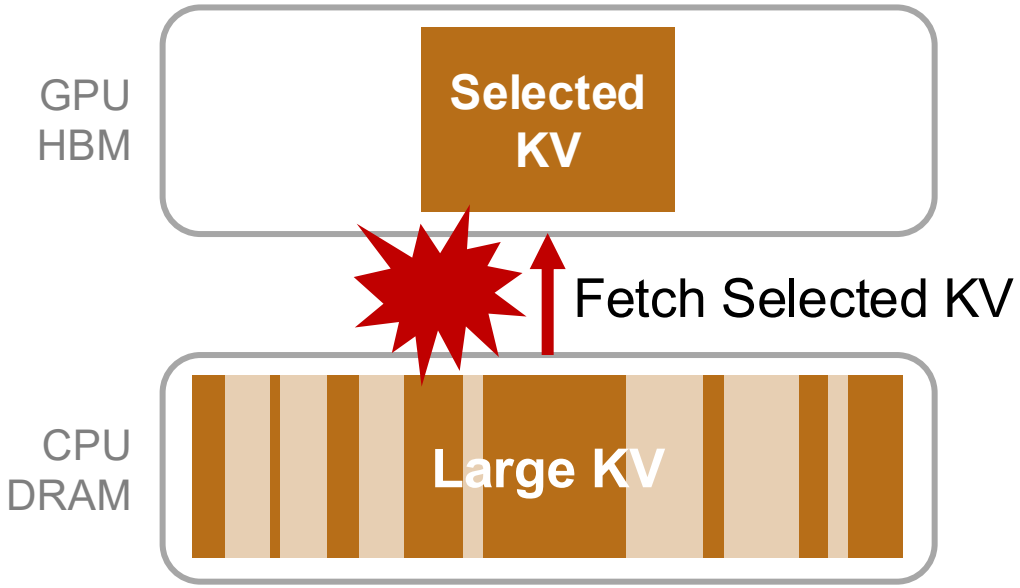
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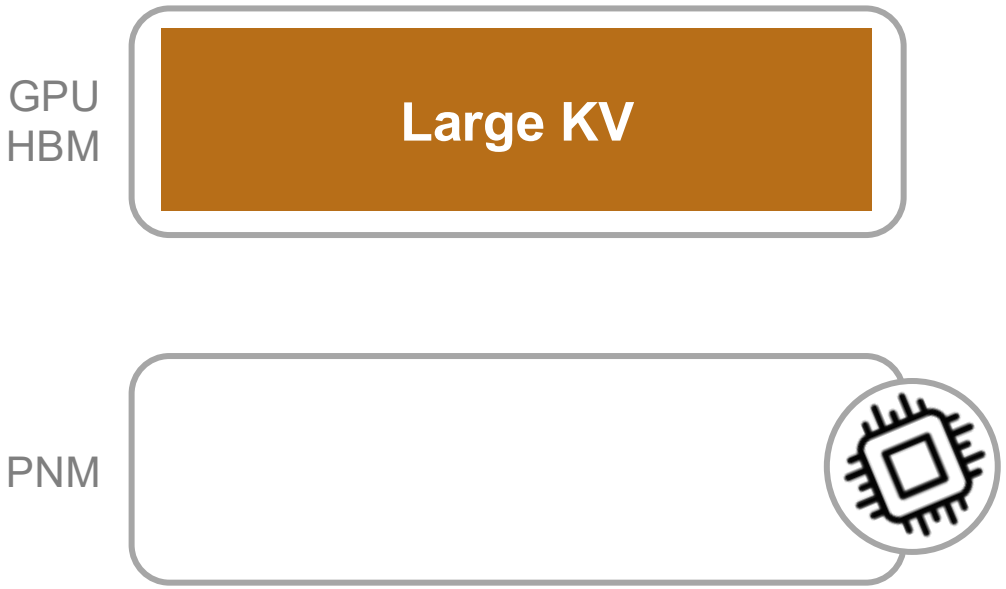
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KV Offloading

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Execute attention where KV resides

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Good fit for long-context decode attention

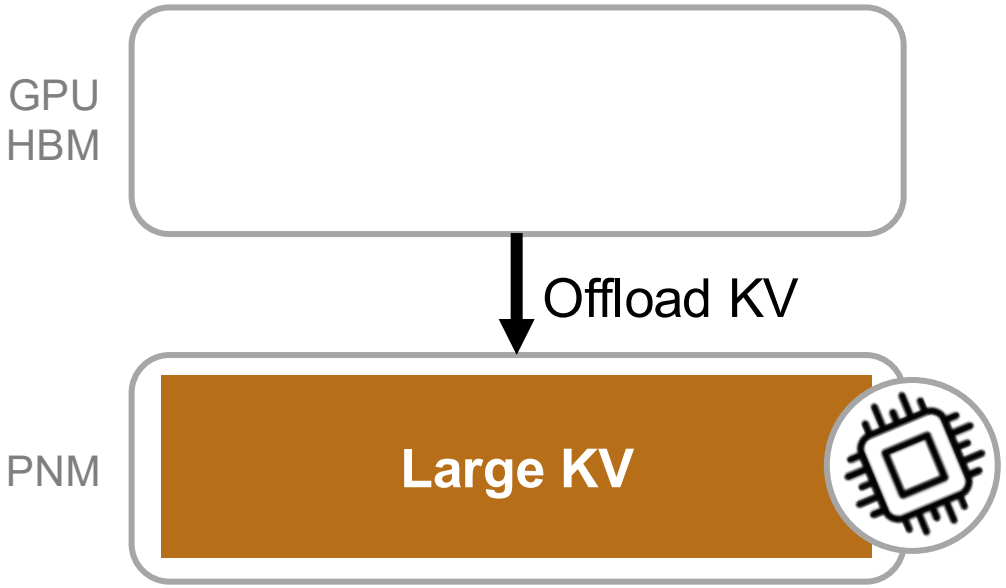
Memory-bound · Large capacity · Sparse execution

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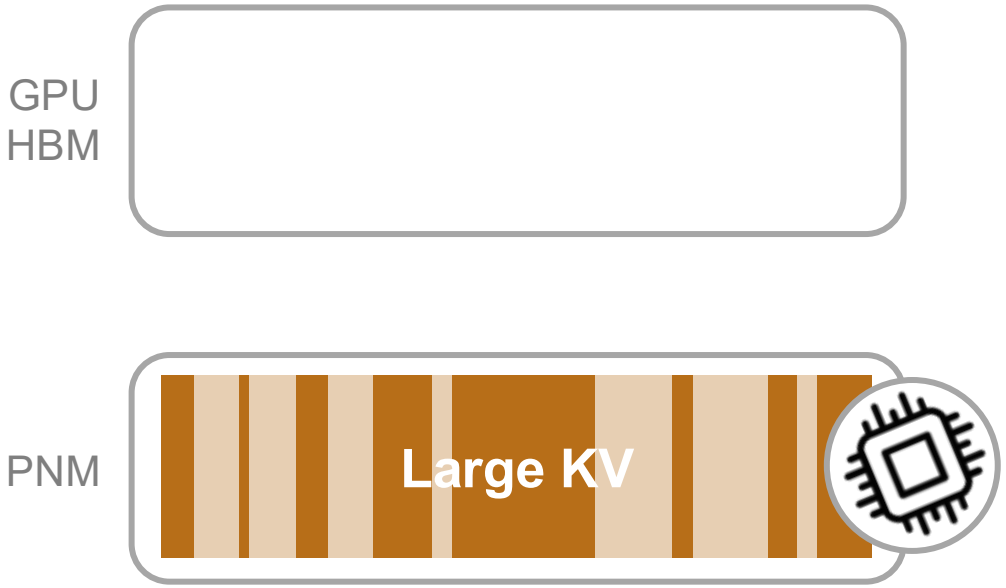
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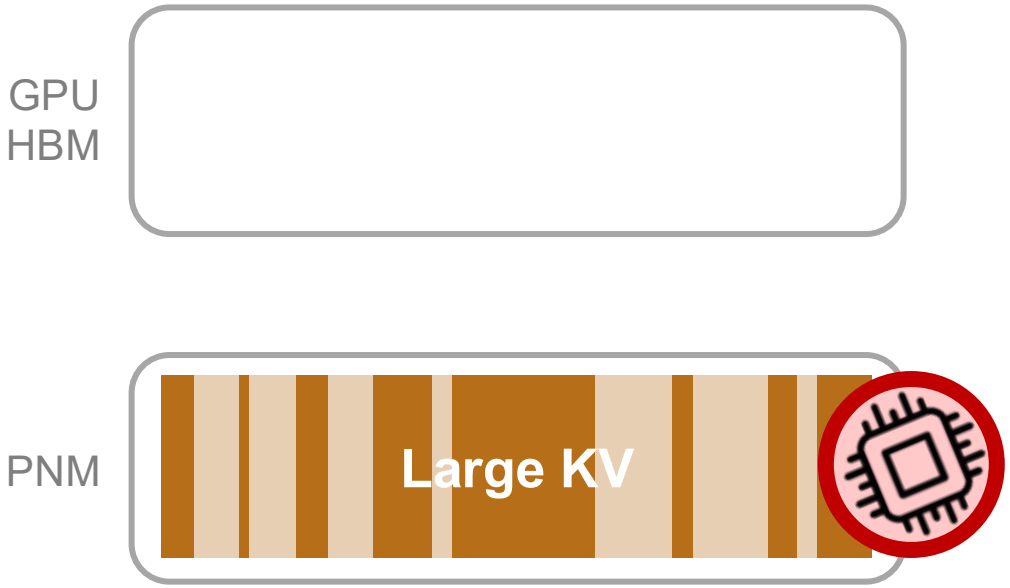
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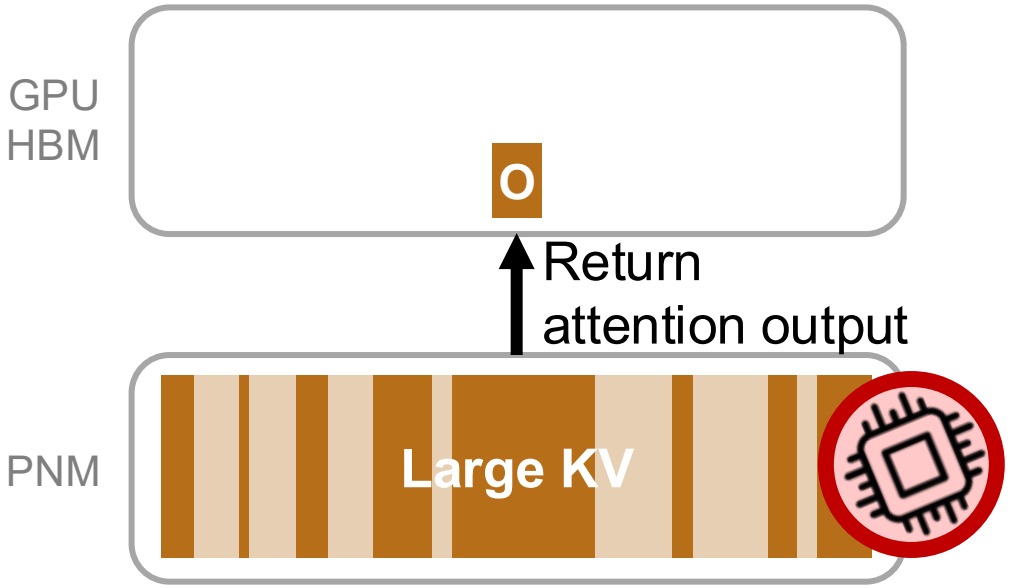
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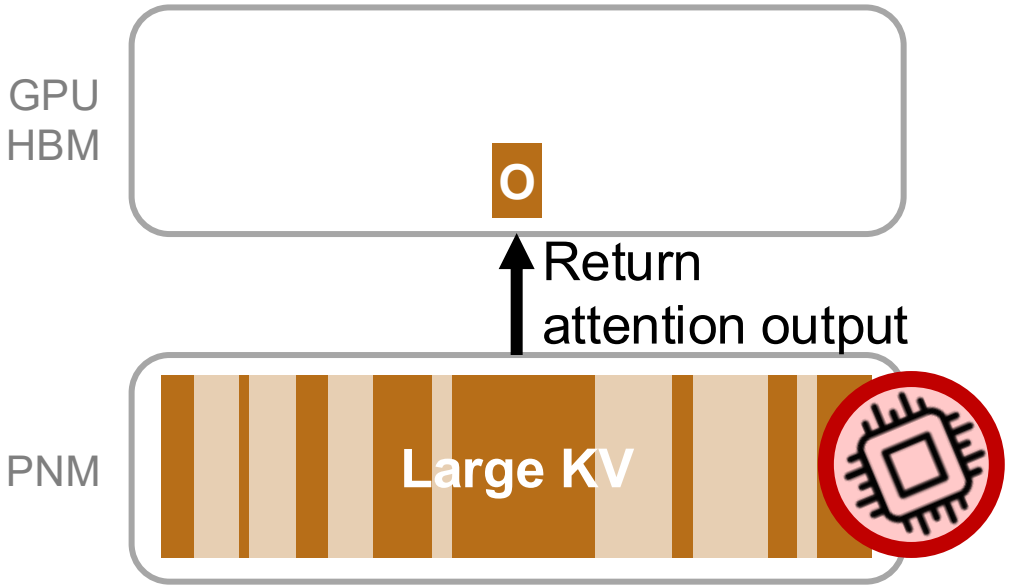
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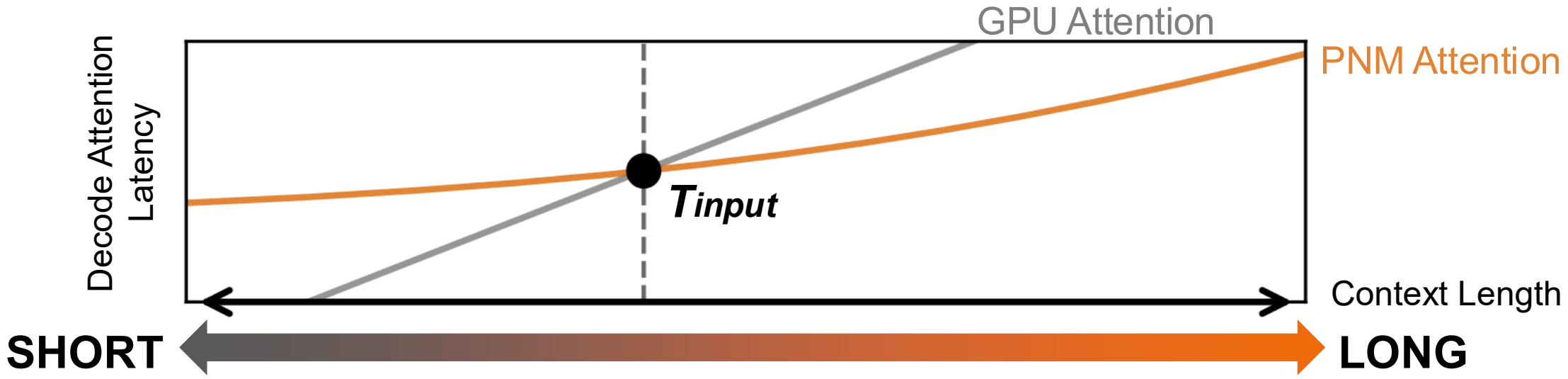
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Memory-bound · Large capacity · Sparse execution

But what about mixed-length workload?

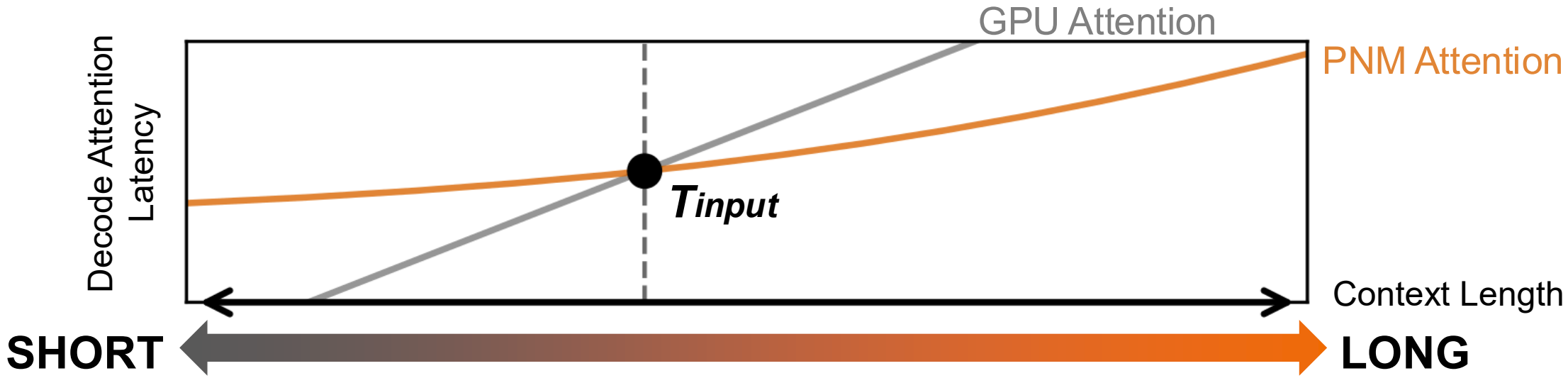
Motivation: Different Lengths, Different Strengths

Observation: GPU and PNM favor different request-length regimes



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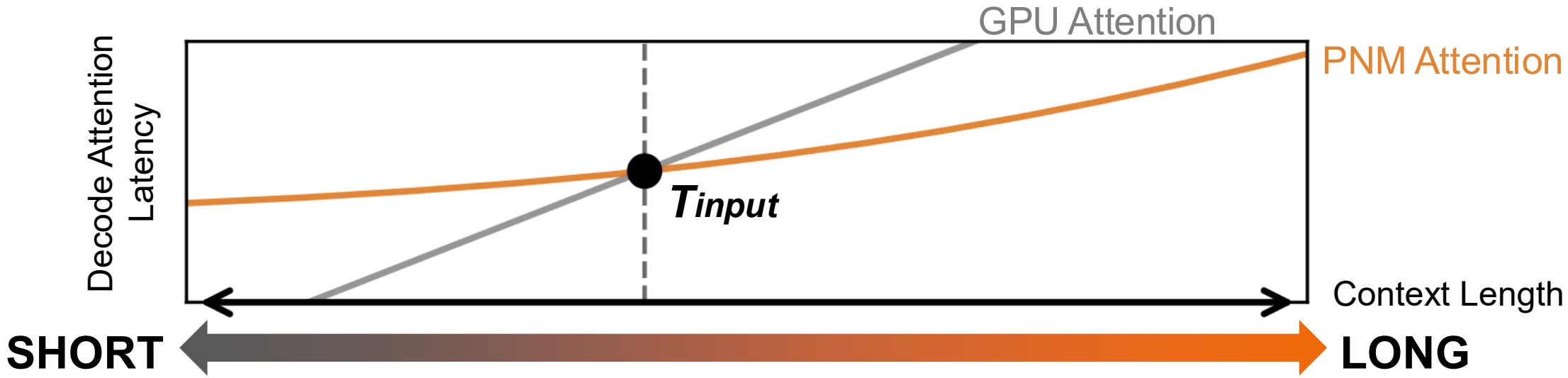


GPU Advantage

- High HBM bandwidth
- No PNM dispatch overhead

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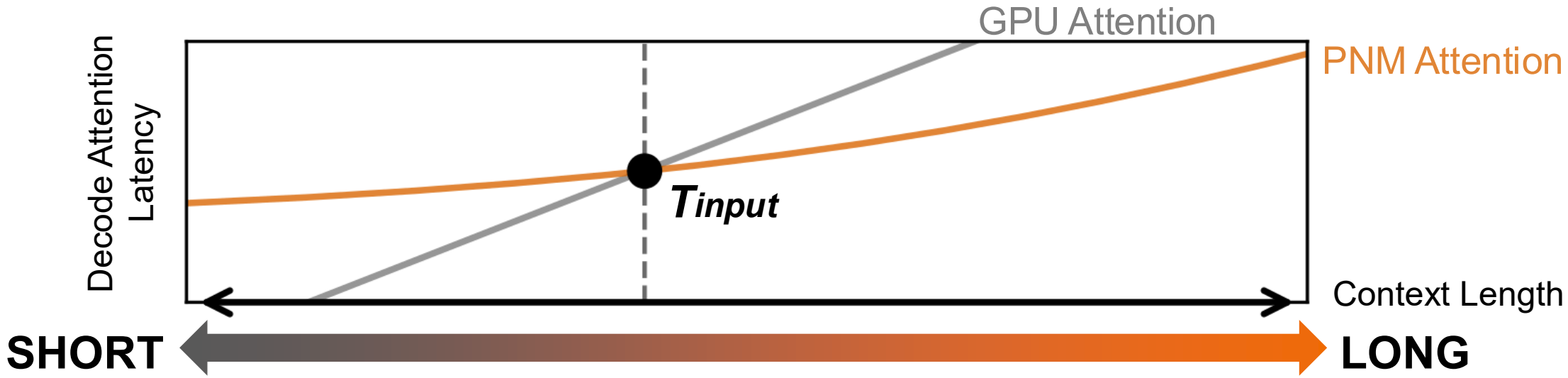
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PNM Advantage

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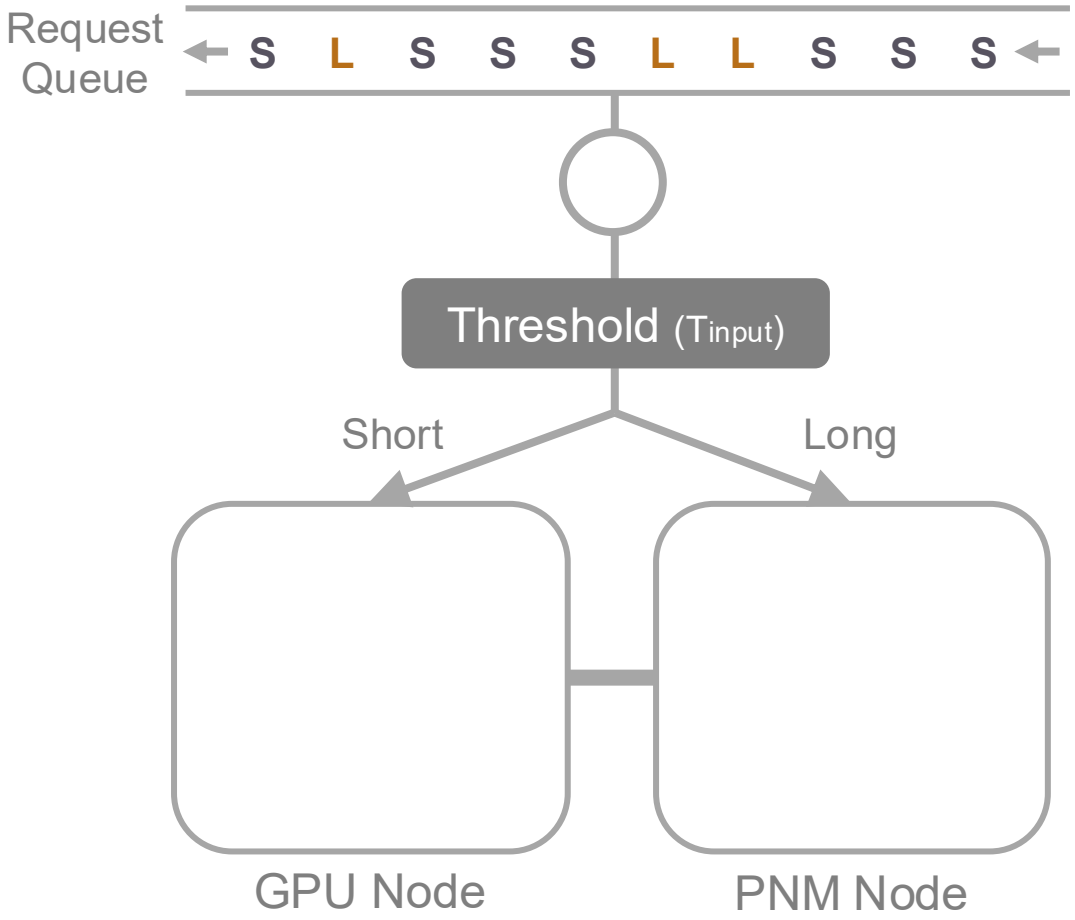
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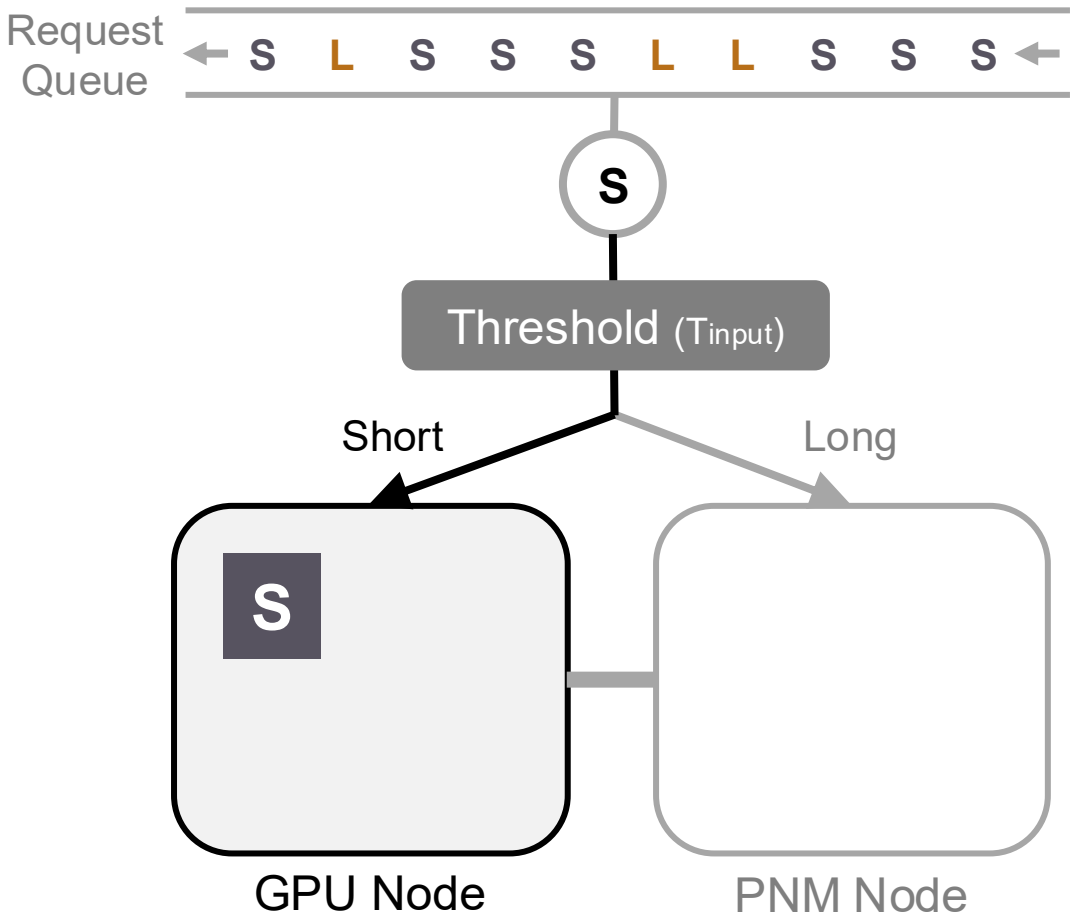
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No single hardware tier is universally superior

NELSSA: Overview

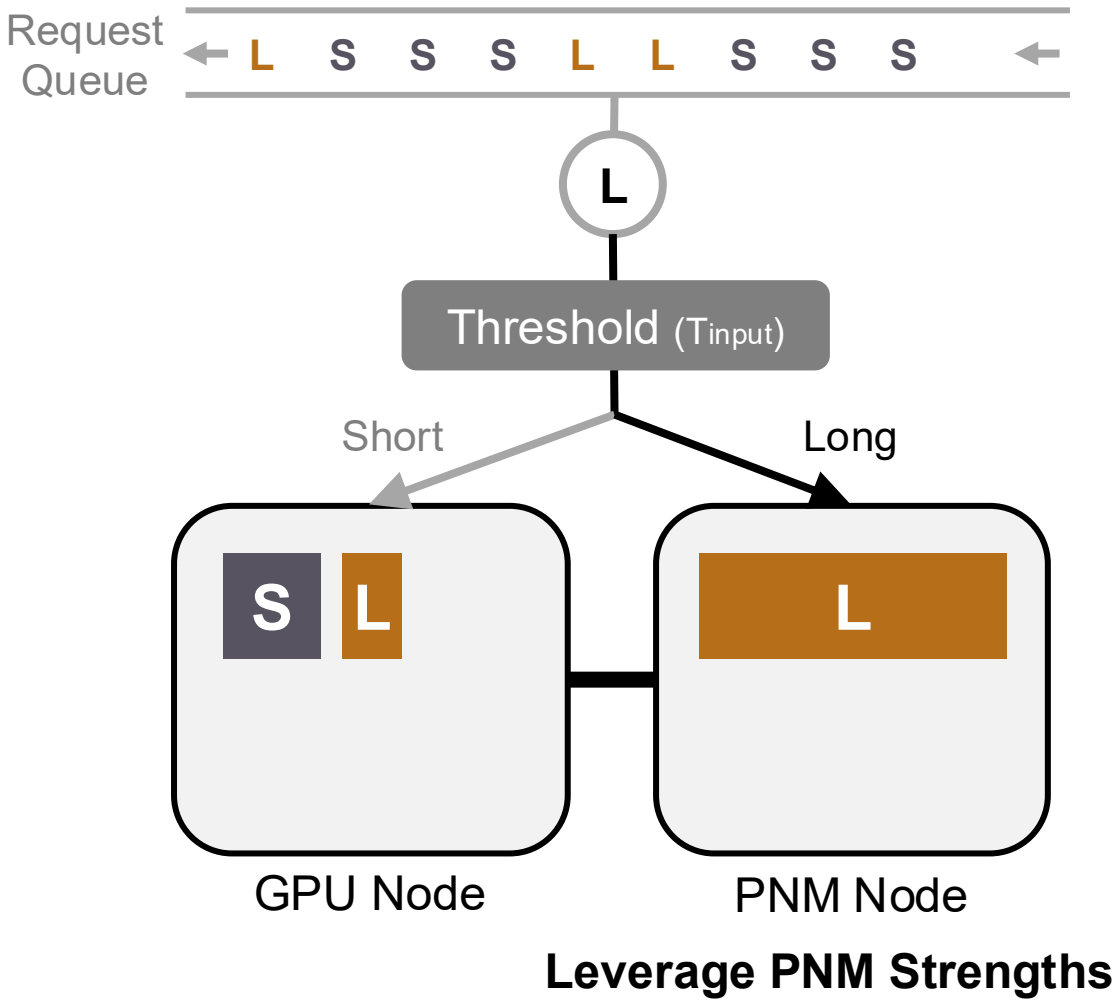


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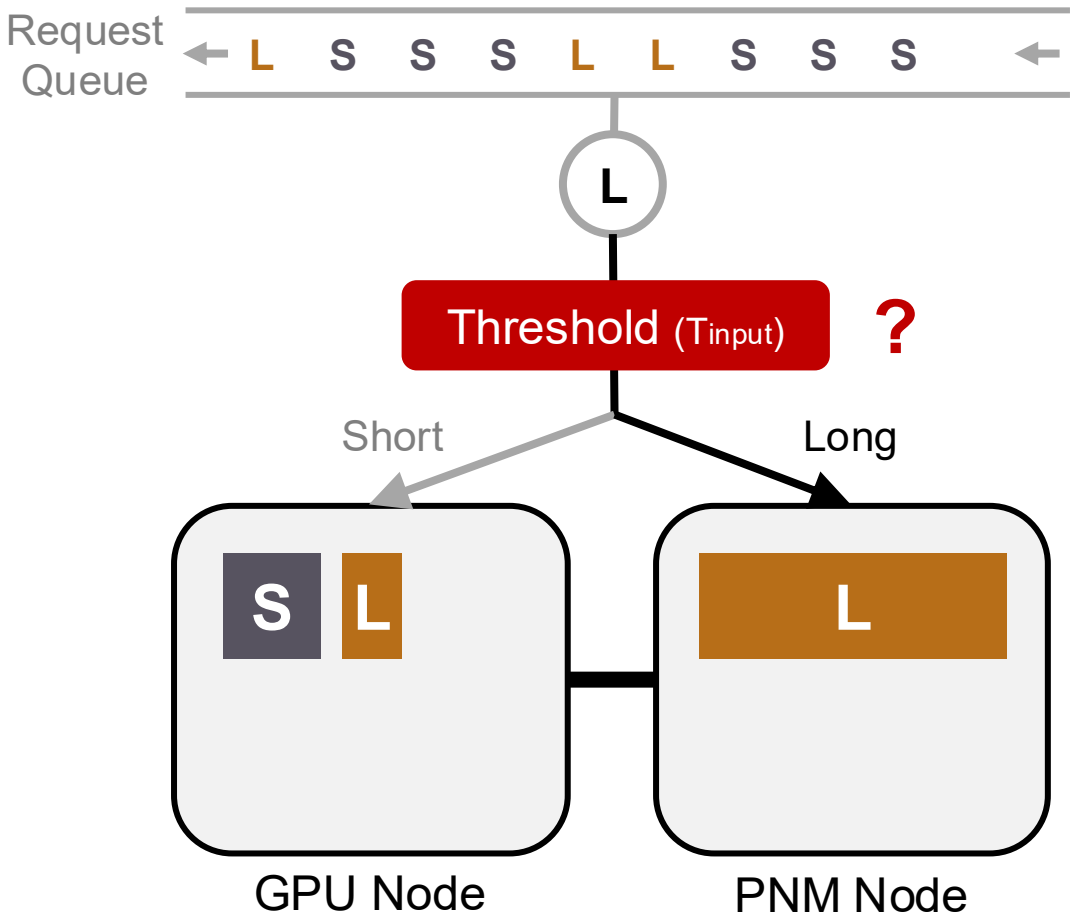
Leverage GPU Strengths

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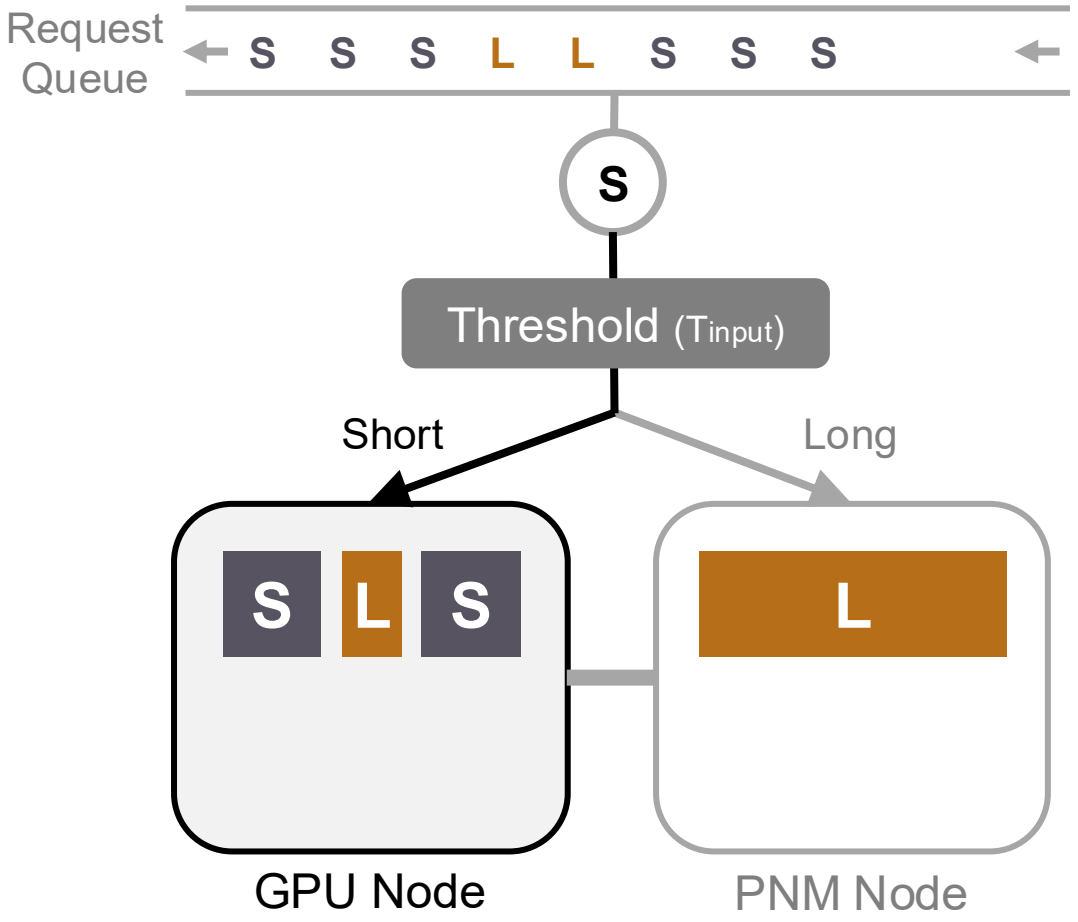
Leverage PNM Strengths

NELSSA: Overview



① **Placement**
How do we determine the routing threshold?

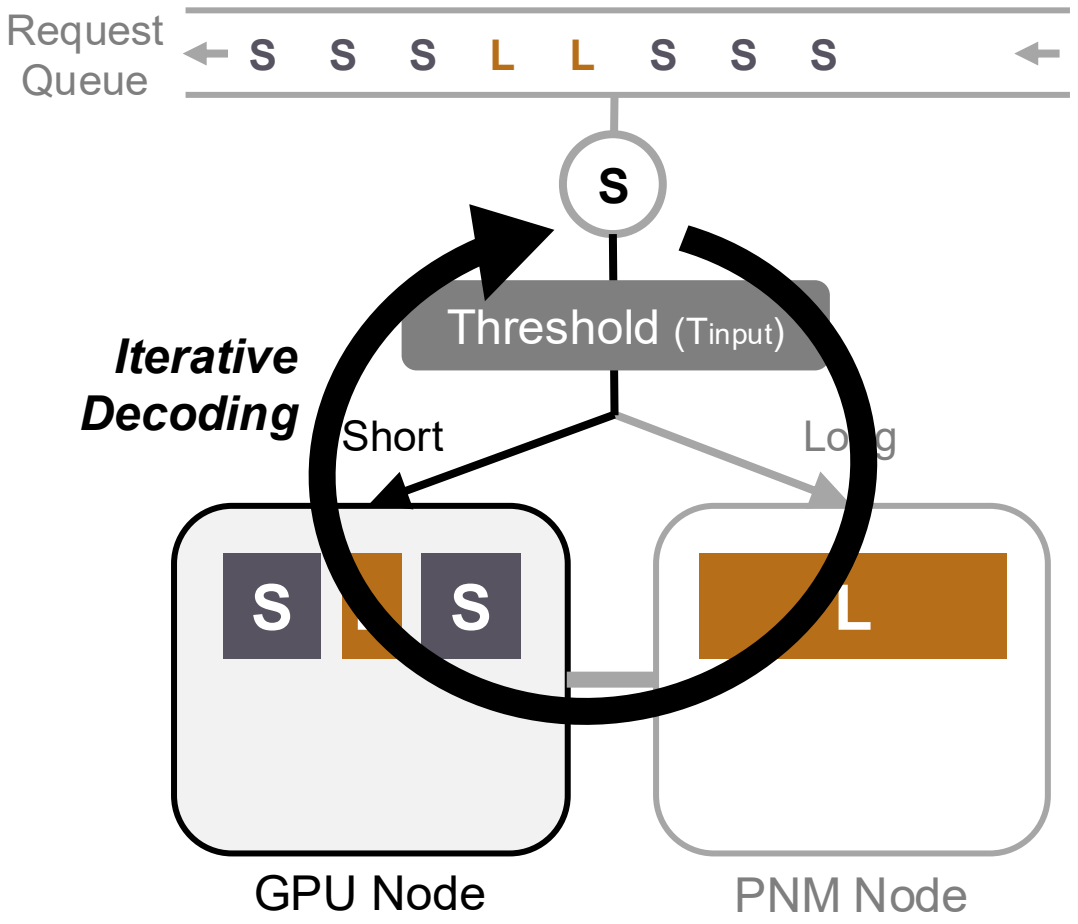
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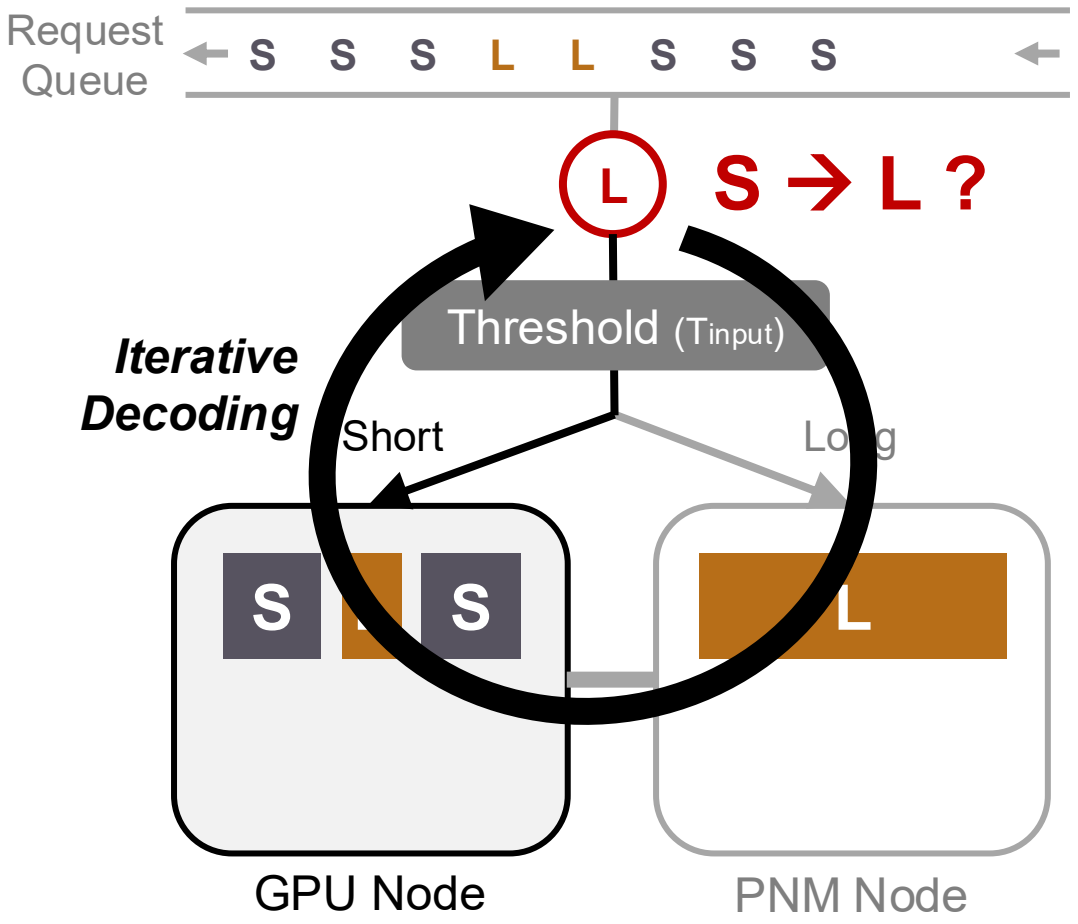
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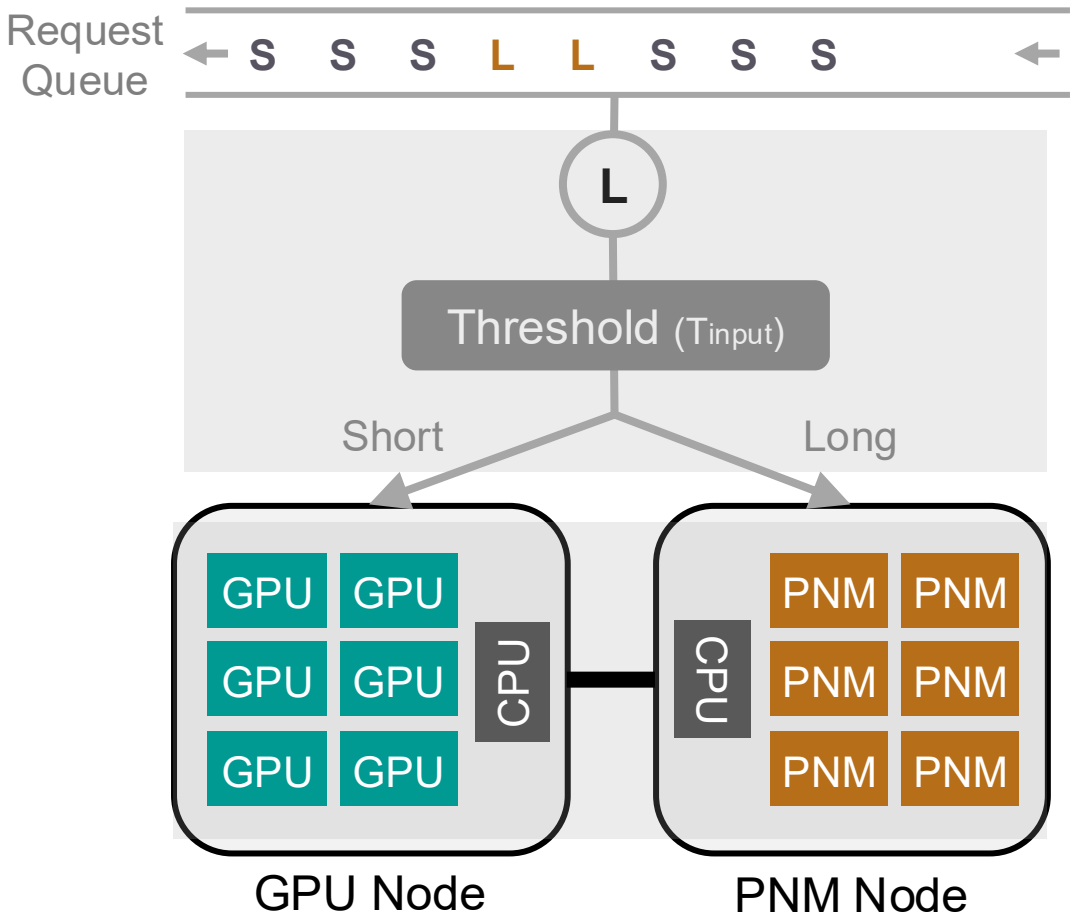
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② **Migration**
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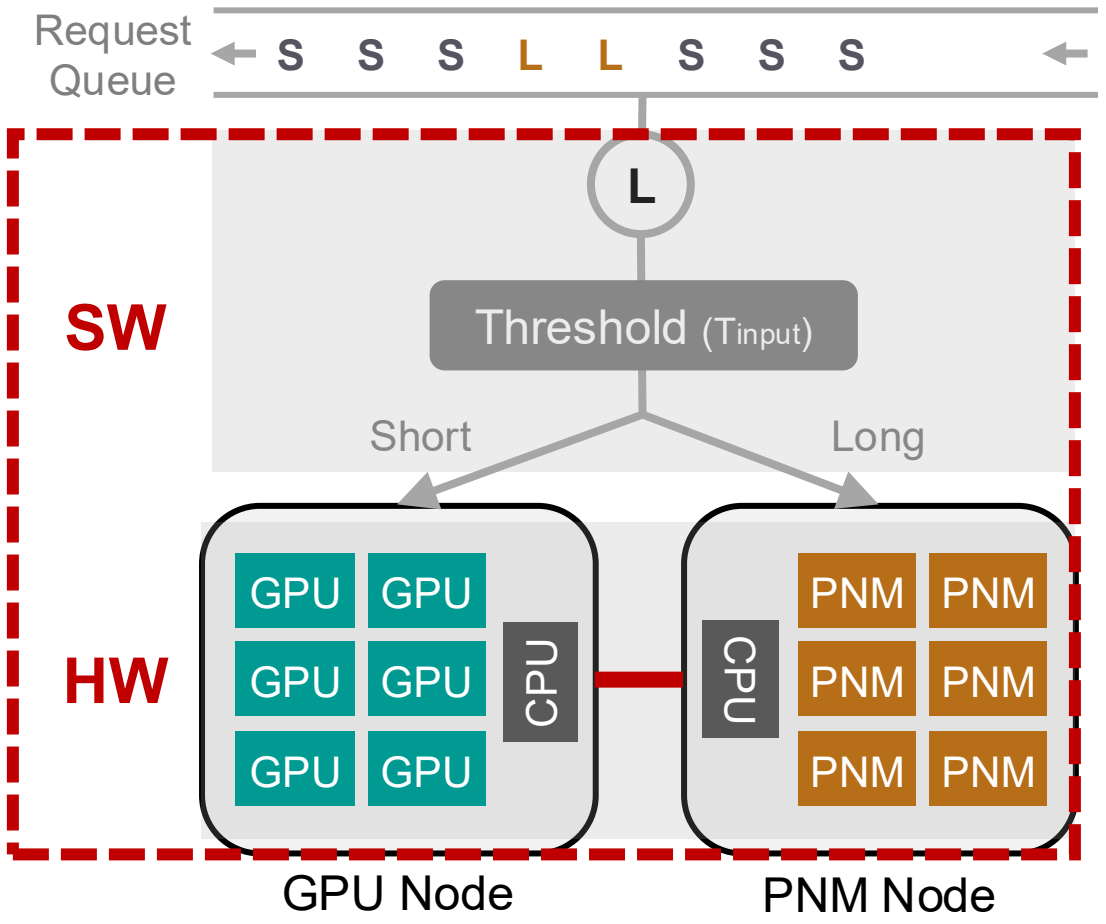
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③ **Integration**
How do we build the end-to-end GPU-PNM system?

NELSSA: Hardware-Aware Routing

Challenge 1

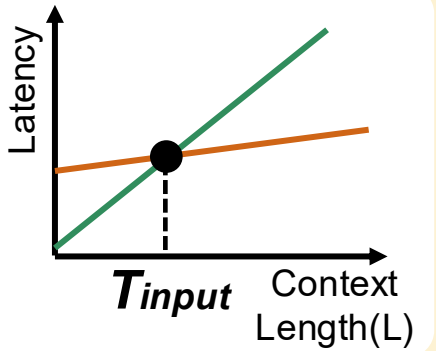
How do we determine the routing threshold?

- Estimate GPU and PNM attention latency from device-level performance characteristics.
- Derive T_{input} as the hardware-specific crossover between the two execution paths.

HBM bandwidth, Compute capability…

GPU attention latency model $T_{GPU}(L)$

PNM attention latency model $T_{PNM}(L)$



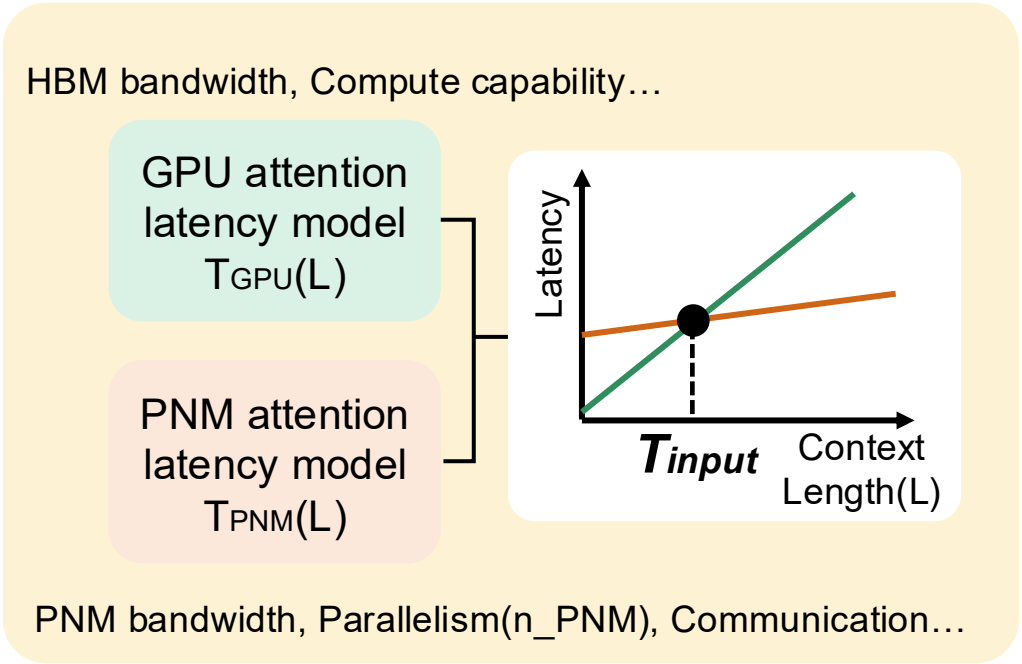
PNM bandwidth, Parallelism(n_PNM), Communication…

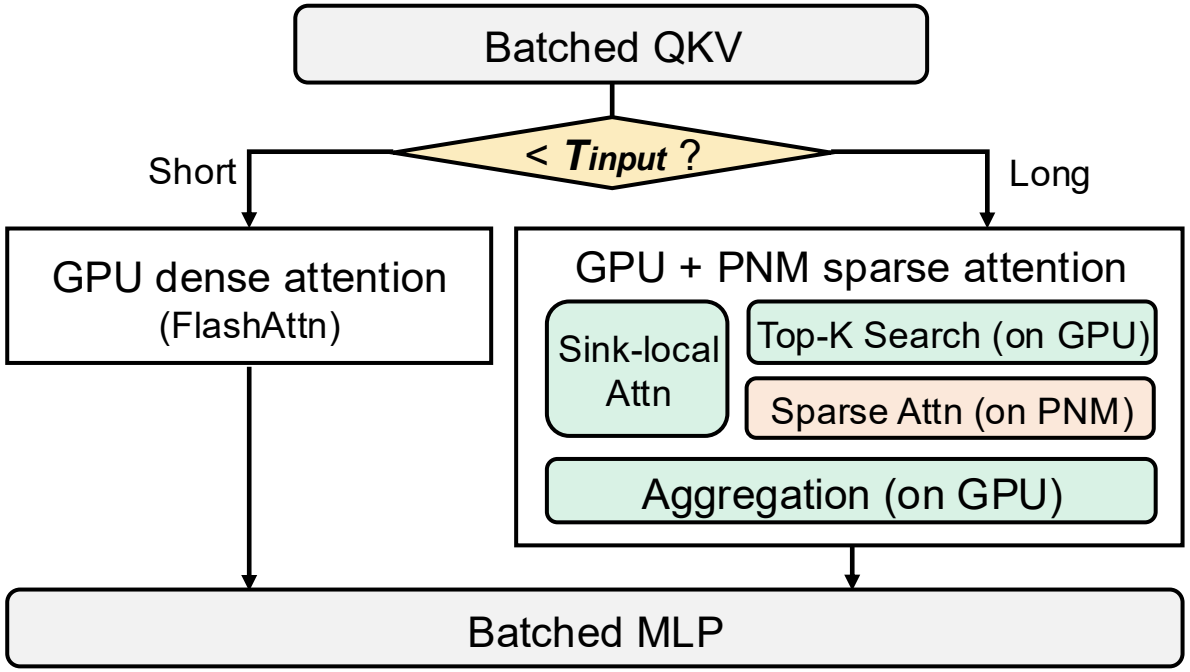
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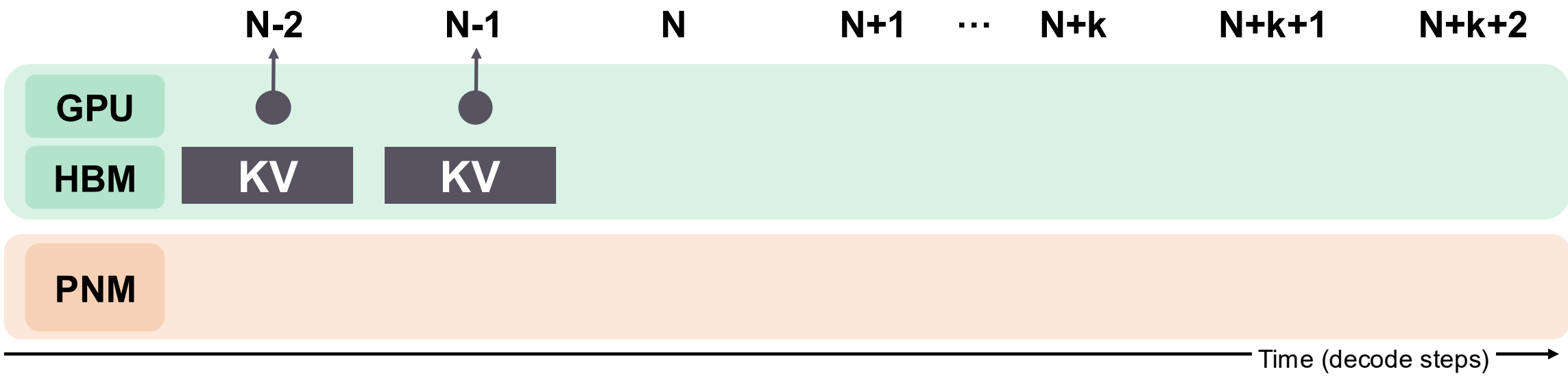




NELSSA: Background Migration

Challenge 2
What if the context grows during decoding?

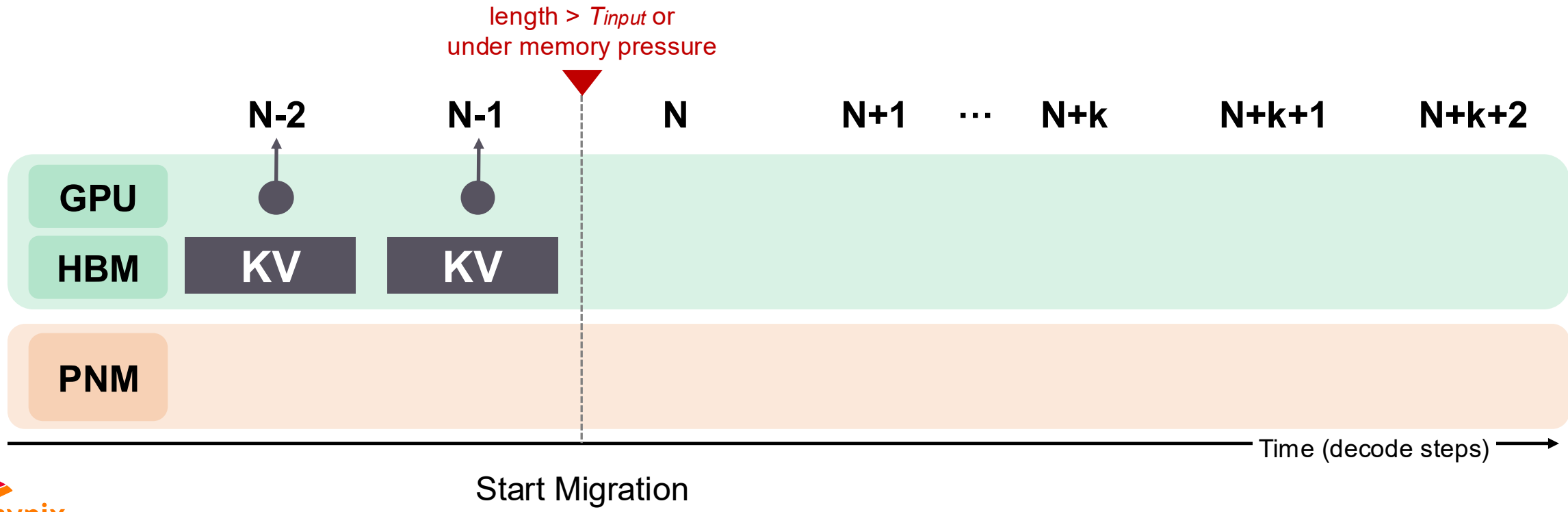
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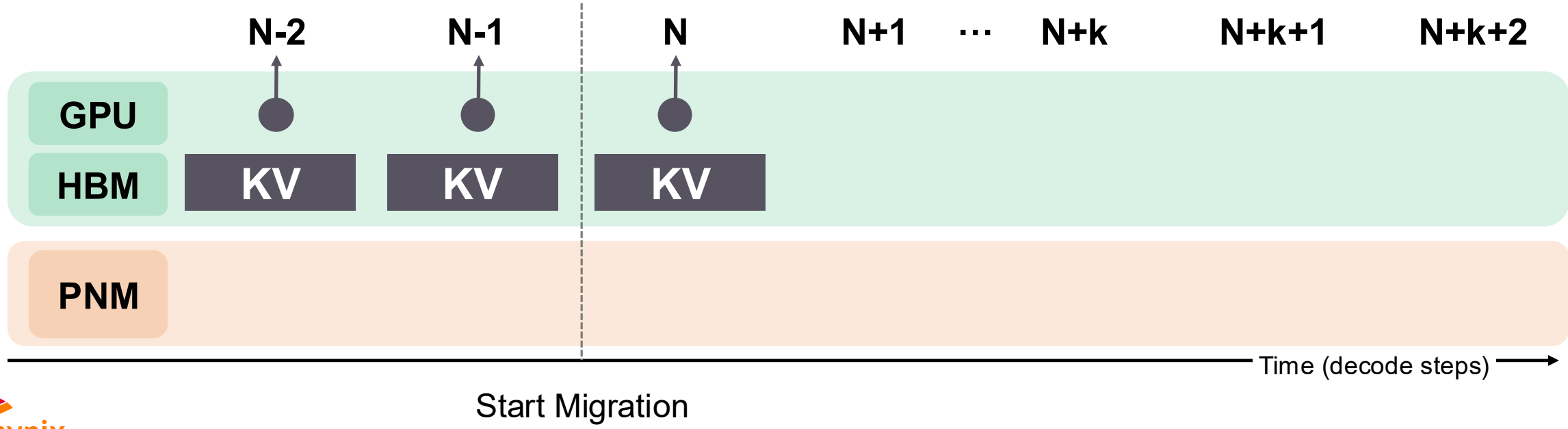
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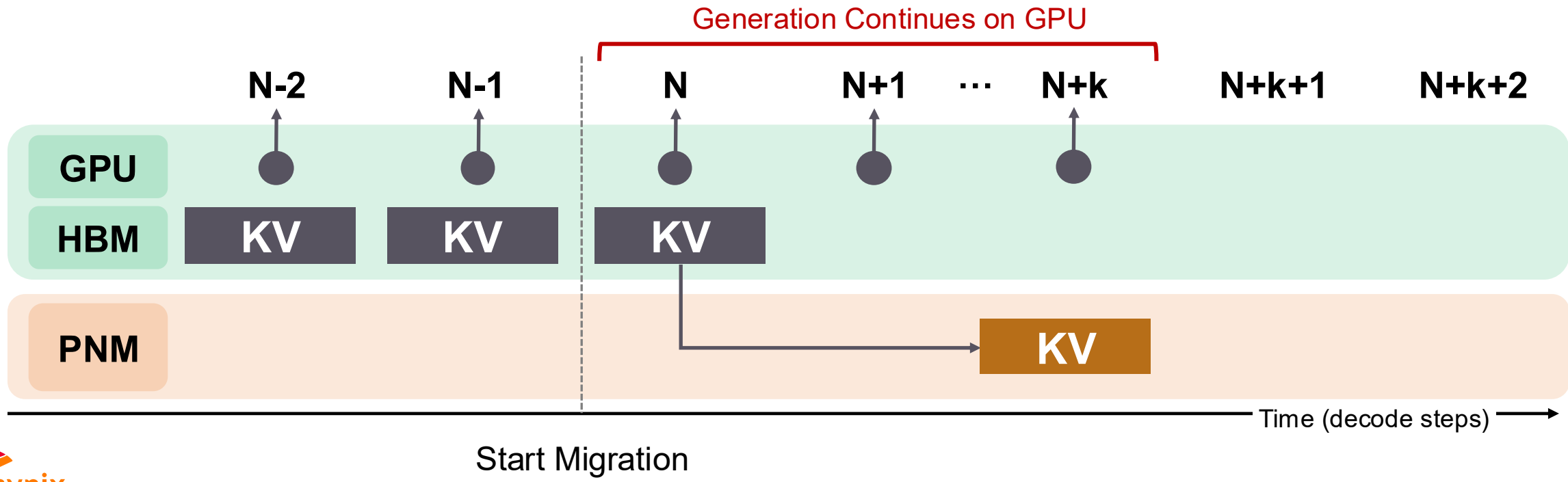
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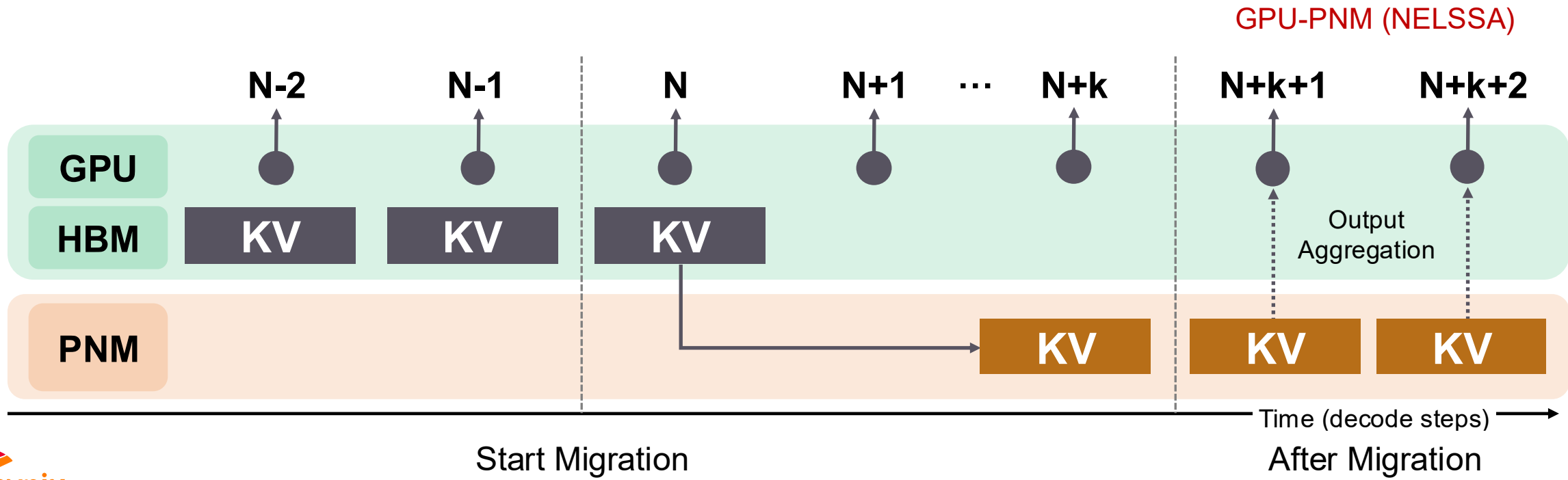
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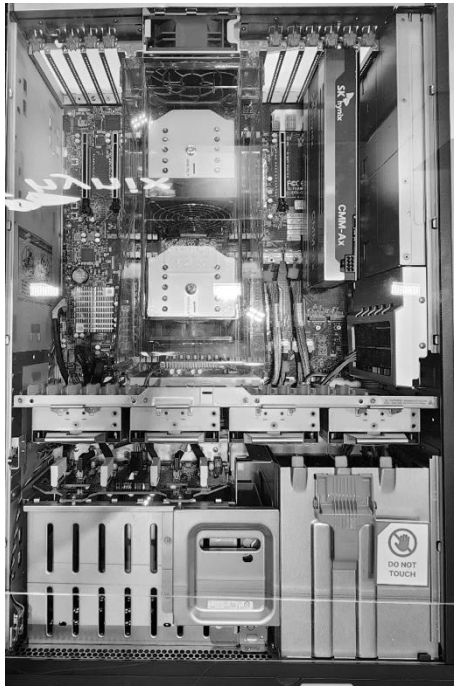


NELSSA: End-to-End System

Challenge 3

How do we build the end-to-end GPU-PNM system?

- NELSSA implements an end-to-end GPU-PNM heterogeneous serving system on real HW
  1) Real HW Prototype: built on real CXL-PNM HW



[CMM-Ax Server Integration]



[FMS 2026 Tech Session]

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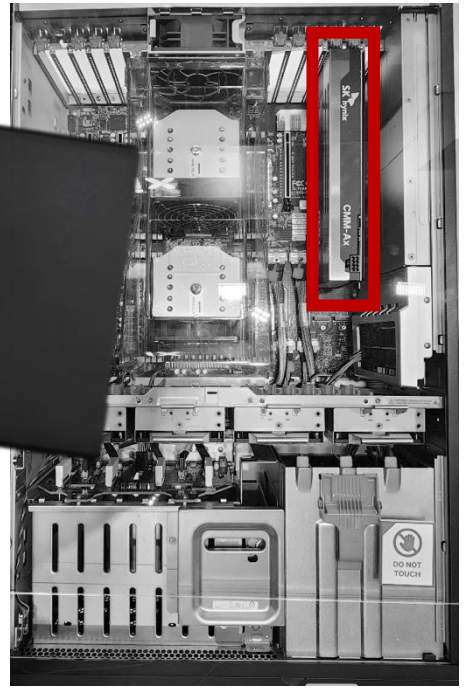
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[SK hynix CMM-Ax]
- CXL 2.0, DDR5-based PNM (32 GB)
- ARM Neoverse V2

Ongoing: Custom board development (512 GB)



[CMM-Ax Server Integration]



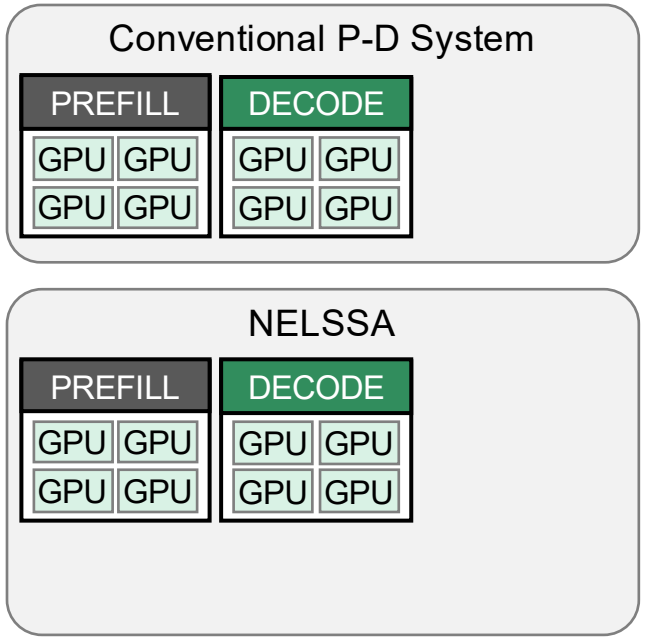
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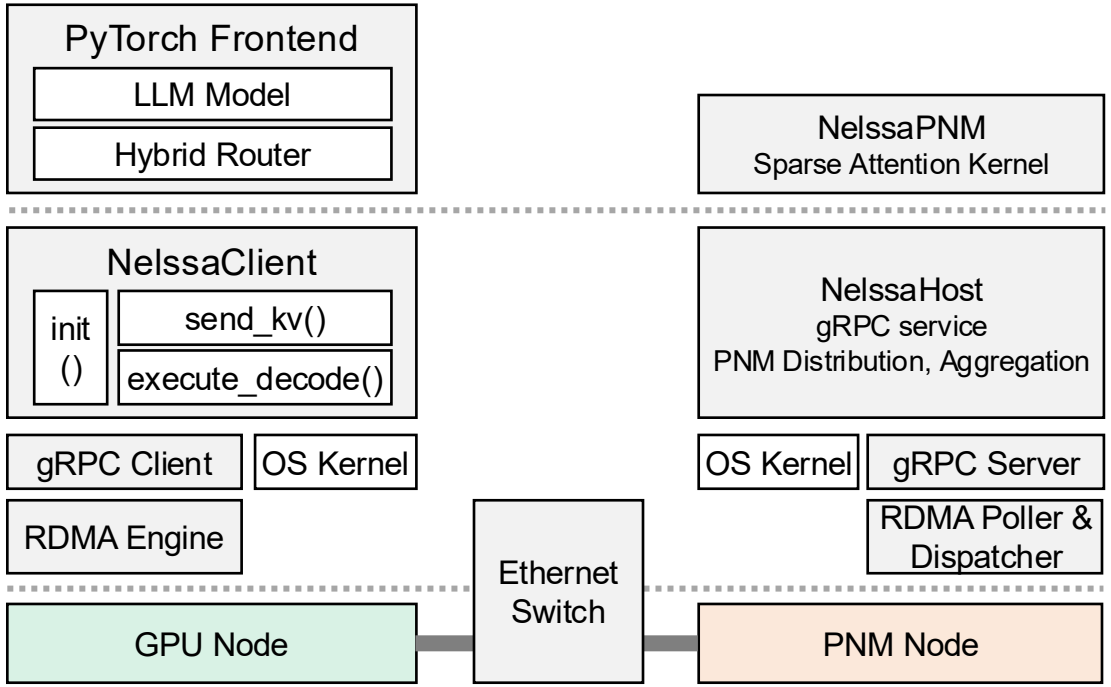
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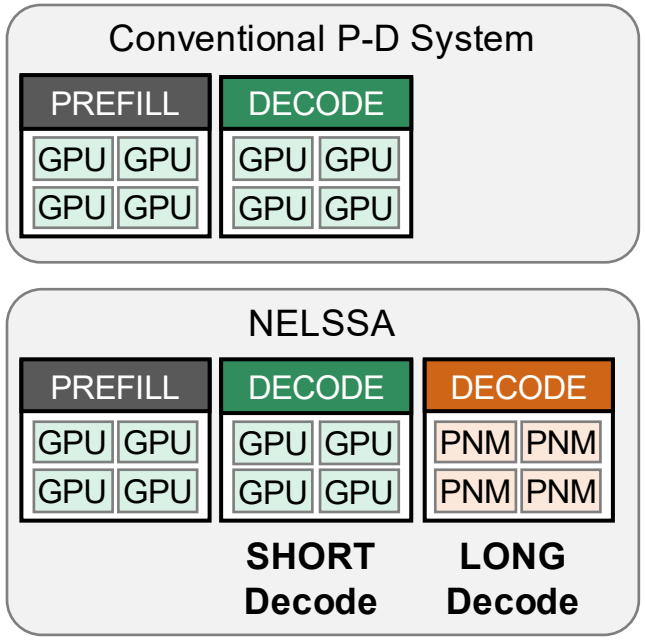


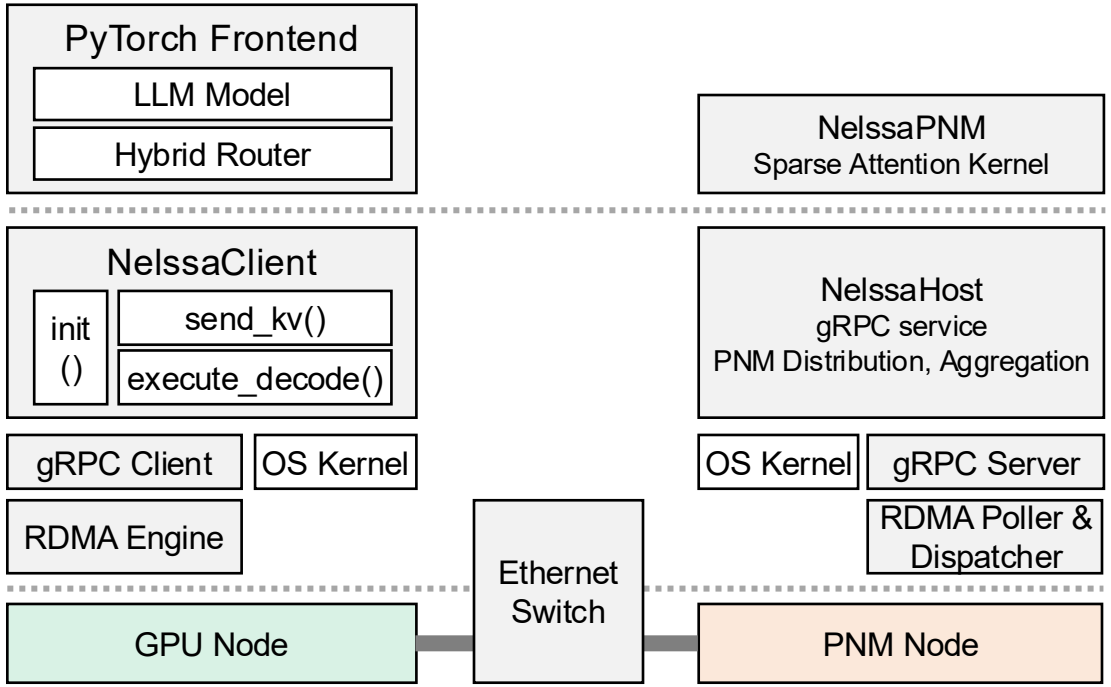
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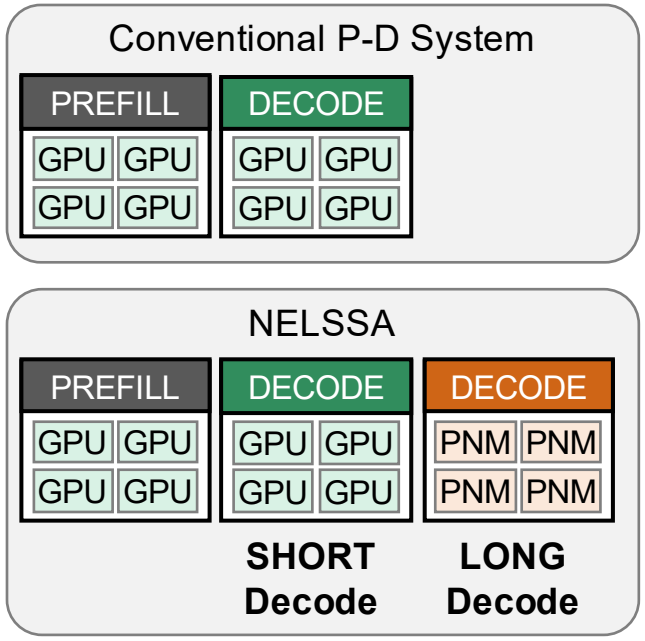


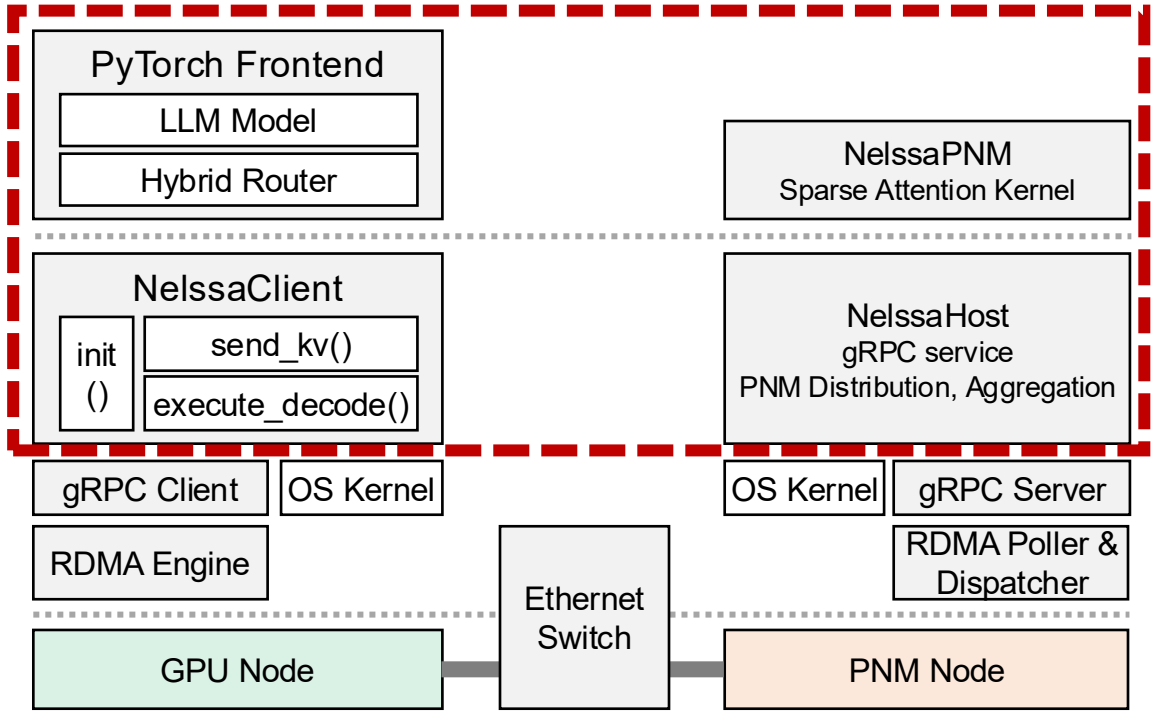
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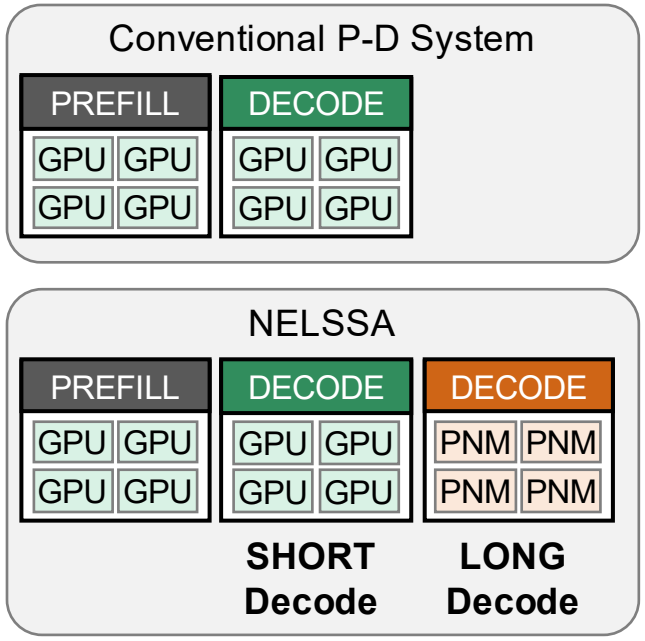


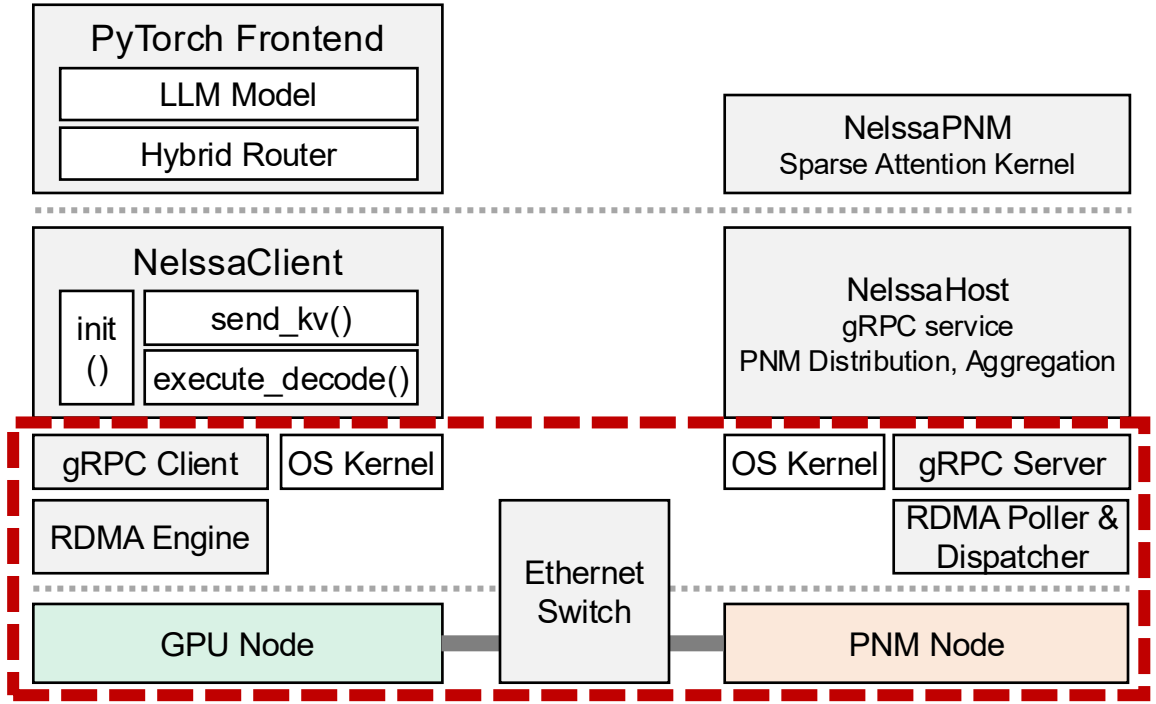
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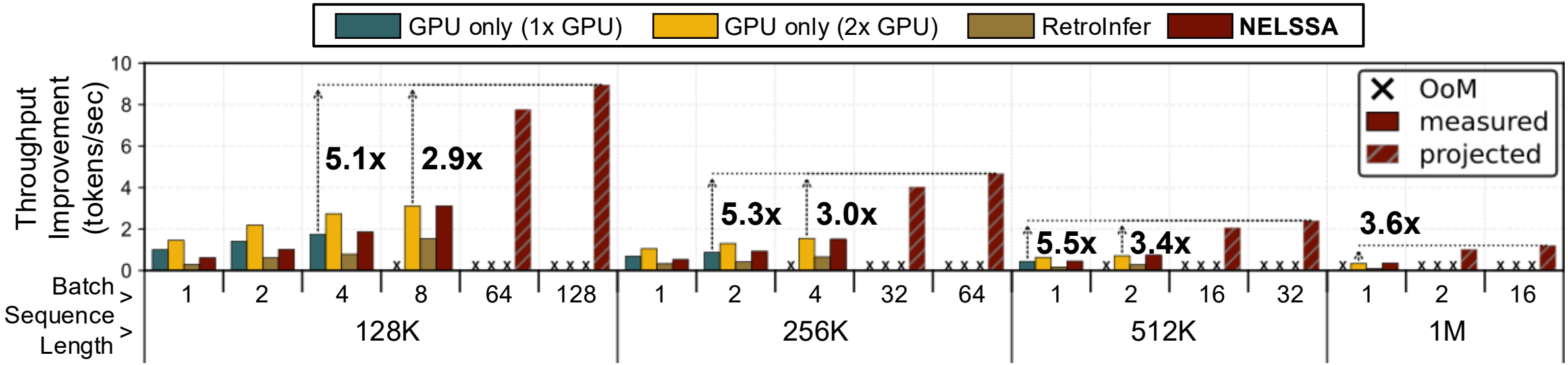




Evaluation: Experimental Methodology

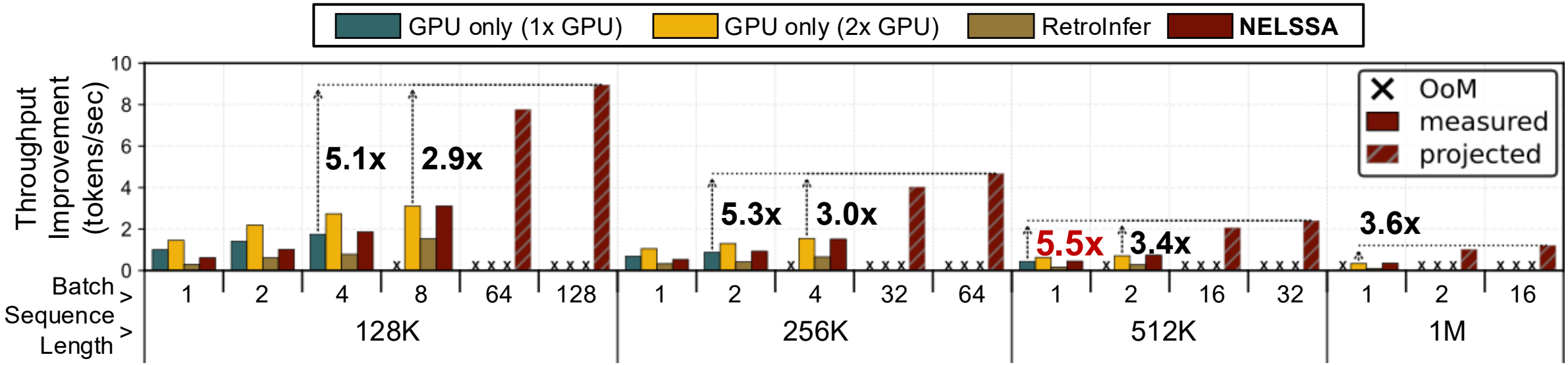
- **Hardware / System**
  - 1x H100 NVL + 4x CXL-PNM devices
  - PNM: 32 GB/device measured, 512 GB/device projected
- **Model / Workload**
  - Llama3-8B-1048K, 128K-1M context
  - RULER for accuracy, Mooncake trace for mixed-length workloads
- **Baselines**
  - 1x GPU: H100 NVL, FlashAttention
  - 2x GPU: 2x H100 NVL, FlashAttention
  - RetroInfer: CPU-offloaded sparse attention
- **Projection**
  - Scale PNM capacity: 32 GB → 512 GB/device
  - Project from measured component-level latency
  - PNM-attention model validated within ±5.6%

Evaluation: Throughput Results



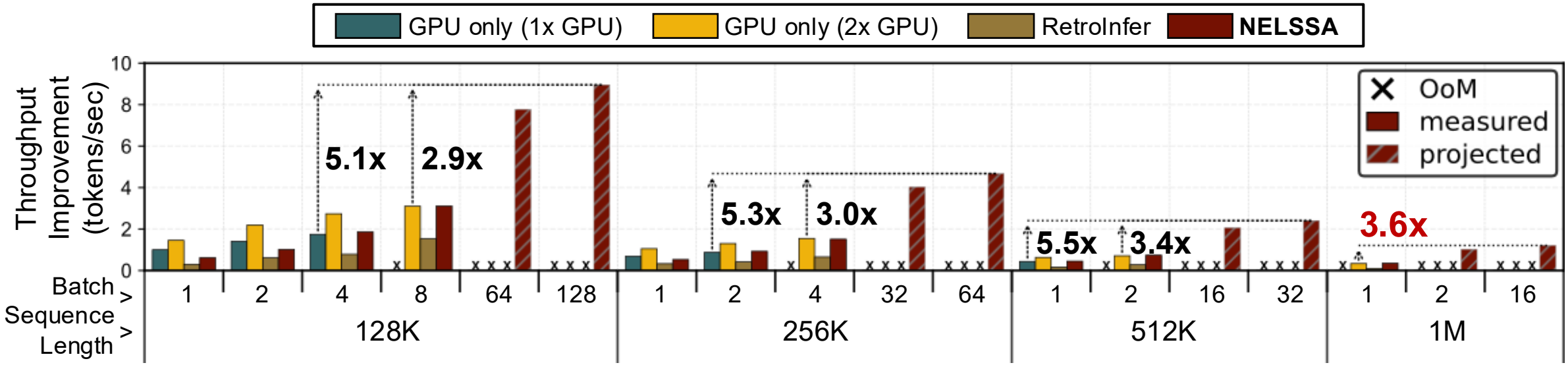
- NELSSA efficiently serves large-KV configurations beyond GPU memory capacity.
- It improves throughput by **up to 5.5× vs. 1× GPU** and **3.6× vs. 2× GPU**.

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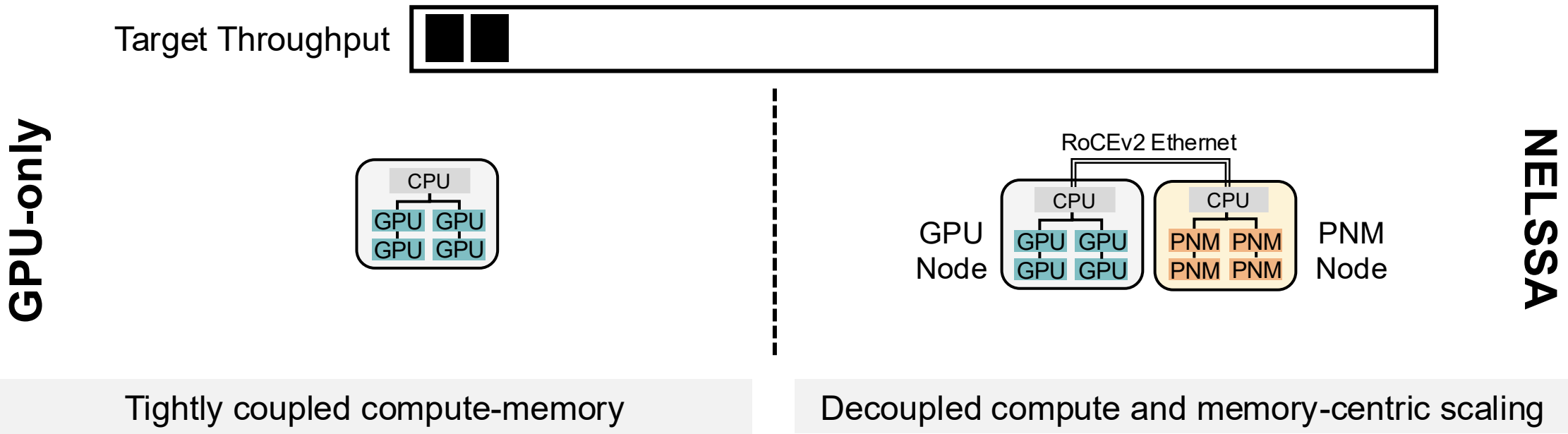
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Evaluation: More Results in the Paper

- **Accuracy-Throughput Trade-off**
  - Stable accuracy at ≥ 4% Selection (RULER benchmark)
- **Mixed-Length Serving**
  - Up to 15x lower P99 latency
- **Long-Request Sensitivity**
  - Robust across varying workload mixes
- **Dynamic Context Growth**
  - 29% higher throughput with background migration
- **System Overheads**
  - Low communication and migration overhead

Implication: NELSSA Scaling

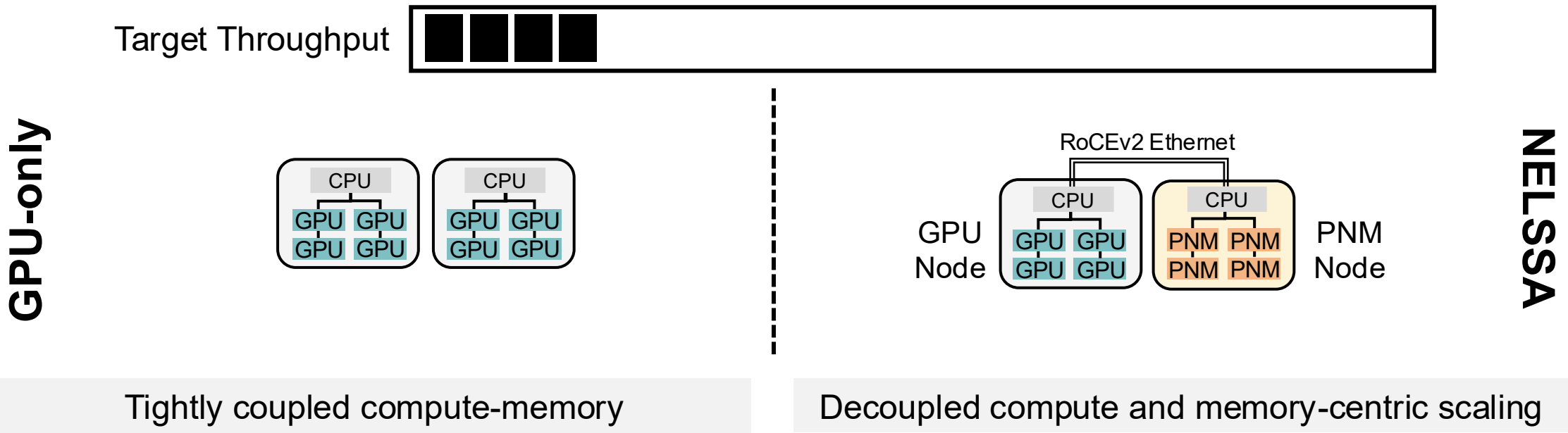
- GPU-PNM disaggregation enables compute and memory-centric to scale independently.
- Resource provisioning can follow workload composition instead of scaling GPUs together.



Illustrative capacity-based scaling example (llama-3-8B)
GPU-only: 4x H100, NELSSA: 4x H100 + 4x PNM (■=1.15M)

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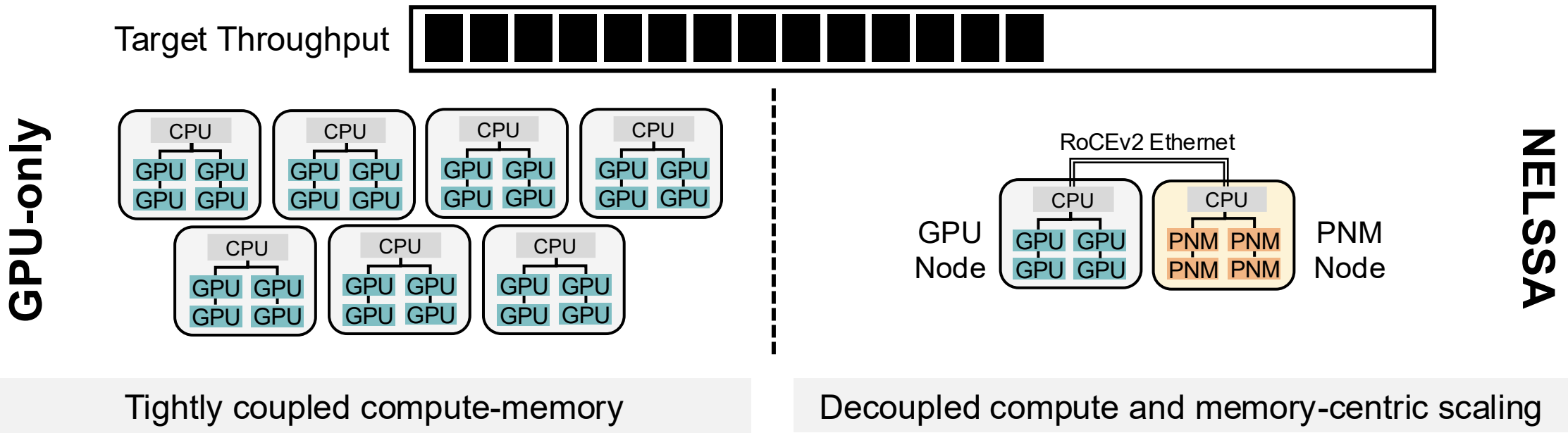
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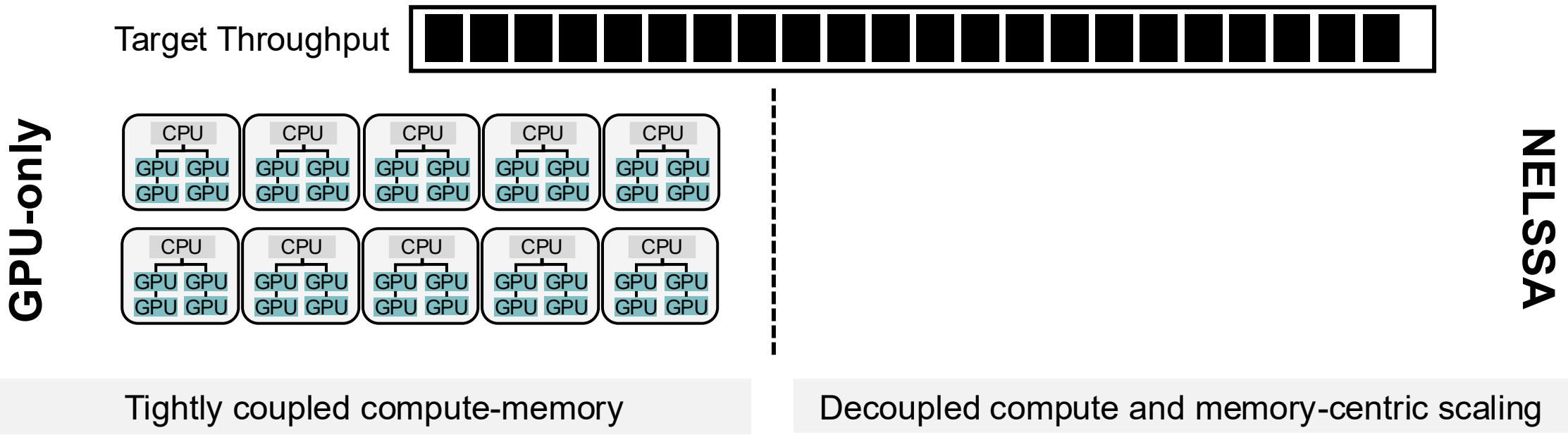
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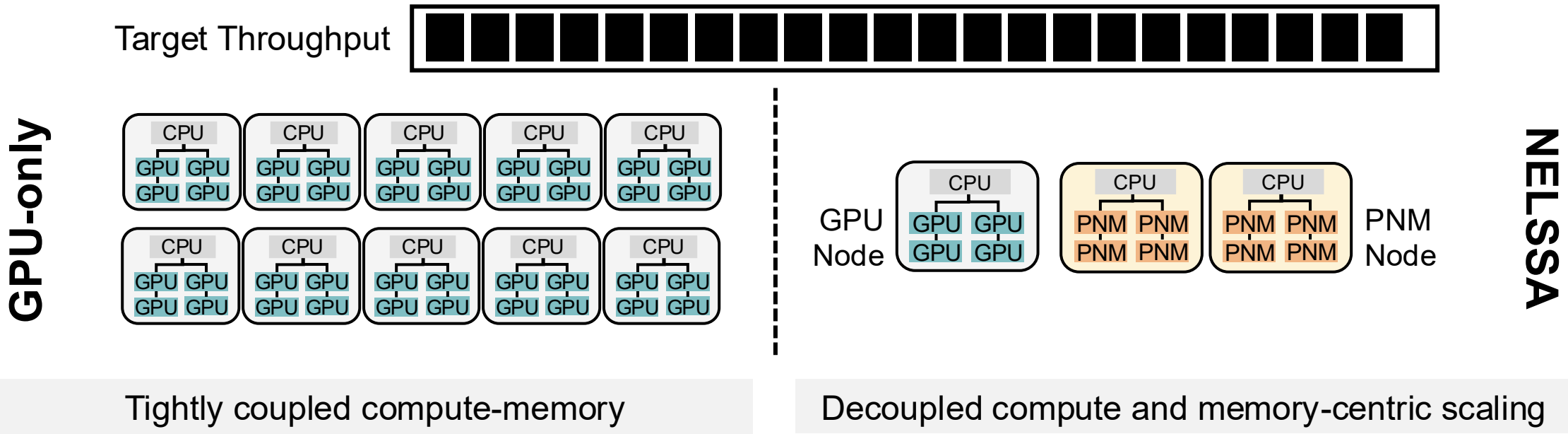
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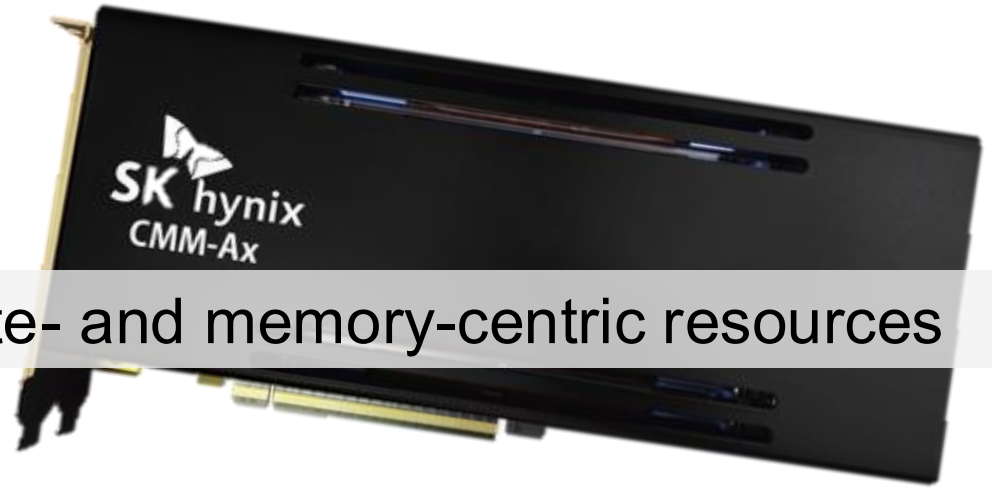
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  - A GPU-PNM heterogeneous system for efficient mixed-length LLM serving
- **Problem**
  - Long requests cause GPU memory pressure and short-request interference, exposing the efficiency limitations of single-tier serving
- **Contributions**
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Toward LLM infrastructure that combines compute- and memory-centric resources



THANK YOU

SK hynix